

BAROCLINIC INSTABILITY

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1. INTRODUCTION

The study of baroclinic instability has its origins in attempts to explain the genesis of midlatitude synoptic storm systems, an endeavor that has been in progress at least since the time of Aristotle¹ [Aristotle, *Meteorologica*, Book III, Chapter 1; see Kutzbach (1979) for the history of the intervening work leading up to the polar front theory in the early part of the present century]. Synoptic eddies are now understood to arise from the release of potential energy stored in the pole-equator temperature gradient of the Earth and similar planets; on a rapidly rotating planet, geostrophic balance links the vertical shear of the wind to this gradient, and the resulting jet systems are referred to as *baroclinic*. Ever since the theoretical work of Charney (1947) and Eady (1949) uncovered the basic mechanism of large-scale instability of a geostrophically balanced baroclinic jet, the field has proceeded in two distinct streams, which sometimes seem about to merge, and at other times seem continents apart. The first stream deals with the linear instability of a specified flow field. Although

¹ Aristotle has somewhat mixed up hurricanes and extratropical cyclones in his discussion (a confusion that persists to the present day in the lay mind). The discussion of whirlwinds in the early part of the brief chapter cited, with some imagination, comes close to identifying cyclones as being due to vortex formation from atmospheric jets, and in this regard is philosophically close to the modern viewpoint.

there is some continuing development, the problems are clear-cut, and the resulting theories are precise, computationally tractable, and mathematically elegant. The basic behavior is by now well understood. Linear theory has seen natural and fruitful extensions into the matter of nonlinear life cycles—the sequence of events tracing the evolution of initially small perturbations through the stage where nonlinearity becomes important, whereafter equilibration (and perhaps ultimately decay) of amplitude sets in. Much remains to be done before this chapter can be closed, but in life-cycle studies the role of linear instability in the genesis of synoptic activity is entirely clear. The second stream of inquiry deals with the matter of what, if anything, such results have to do with the observed midlatitude synoptic eddies, and here the state of affairs is rather less satisfactory. Difficulties arise because synoptic eddies in Nature rarely proceed from small perturbations of a nearly undisturbed jet. The two streams come together in the study of the statistical equilibrium behavior of forced/dissipated baroclinic systems (“baroclinic turbulence”), and at some point such models shade over into full-blown atmospheric general circulation models. At that point it is difficult to see what is left of the original linear instability theory. Although we do not lay this matter to rest in the course of our review, we hope at least to clarify the terms of the debate.

Because it is not possible to cover in comprehensive fashion even the topics of current interest in this vast subject, we are content to provide an account of the fundamental material necessary to approach the subject and flesh out some neglected and unfamiliar aspects of the linear theory. Essential observational background is provided in Section 2, and the mathematical basics are laid out in Section 3. There are some notable omissions. Our review for the most part focuses on quasigeostrophic dynamics. We consider only dry dynamics, though latent heat release undoubtedly plays at least a modifying role in terrestrial synoptic eddies. Nor do we cover anything in the elegant field of weakly nonlinear dynamics [cf Pedlosky (1970) and the many related papers cited in Pedlosky (1979)], which is nonetheless valuable in providing an island of certainty in a bewildering sea of nonlinear behaviors. Our discussion is slanted towards the terrestrial atmosphere which is the best-observed natural example we have of a baroclinically unstable system. The applicability of the key concepts to oceans and to other planets—where similar issues are playing out—will hopefully be evident.

Stability criteria are treated in Section 4, and normal modes are discussed in Section 5. Aspects of the linear initial value problem are taken up in Section 6. In Section 7 we touch (though only lightly) on the difficult subject of nonlinear equilibration. Finally, in Section 8, we take stock of where the subject stands and where it is going.

2. OBSERVATIONAL BACKGROUND

Baroclinic instability would be mostly an intellectual curiosity were it not for its perceived connection with the synoptic transient motions of the Earth's atmosphere and related eddies in the Earth's ocean and in the atmospheres of other planets. Synoptic transients are the mobile high/low pressure systems which are responsible for much of ordinary midlatitude "weather," and which make limited weather prediction by barometer possible. Unlike hurricanes and tropical storms, synoptic eddies are not rare or sporadic events. A steady succession of them appears in midlatitude bands in each hemisphere throughout the year.

Synoptic eddies are governed by a dispersion relation of sorts. *Time filtering* of an observed meteorological field to isolate the contributions of narrow frequency bands at each point of space also yields a coherent *spatial* pattern with well-defined spatial scales. Bandpass studies of this type were pioneered by Blackmon et al (1977). In Figure 1 we show a bandpass 200 mb geopotential height analysis, recomputed from the US National Meteorological Center dataset described in Blackmon et al (1977); in midlatitudes, the height field is approximately a streamfunction for the bandpass wind field. Although the 2–5-day band yields structures that agree with the customary subjective picture of synoptic eddies, the atmosphere does not actually have a sharp spectral peak in this band (Blackmon 1976). Rather, the synoptic eddies shade gradually into Rossby waves and other large-scale features of the low frequency variability as lower frequencies are employed. On the other hand, the spatial structure shows little sensitivity to frequency for time scales of less than 10 days (Blackmon et al 1984a,b).

From bandpass and related analyses the following additional features emerge. Except where otherwise noted, these results are for the northern hemisphere winter.

1. The typical zonal scale of a high-low system is 4000 km. Phase speeds are comparable to the time-mean 700 mb wind (10–20 m/s, depending on location). The level where the mean wind equals the phase speed is known as the *steering level*. The eddies at 200 mb propagate much more slowly than the typical 30–40 m/s winds prevailing at those altitudes (cf Figure 1), which constitutes strong evidence that the eddies are driven from below. Crude estimates of growth rates based on unfiltered data are suggestive of a doubling time of 1–2 days aloft (cf Figures 2, 3, 5, and 6 of Chang 1993).
2. The typical meridional scale of an individual low or high pressure cell is roughly equal to the zonal scale of the cell, and the meridional

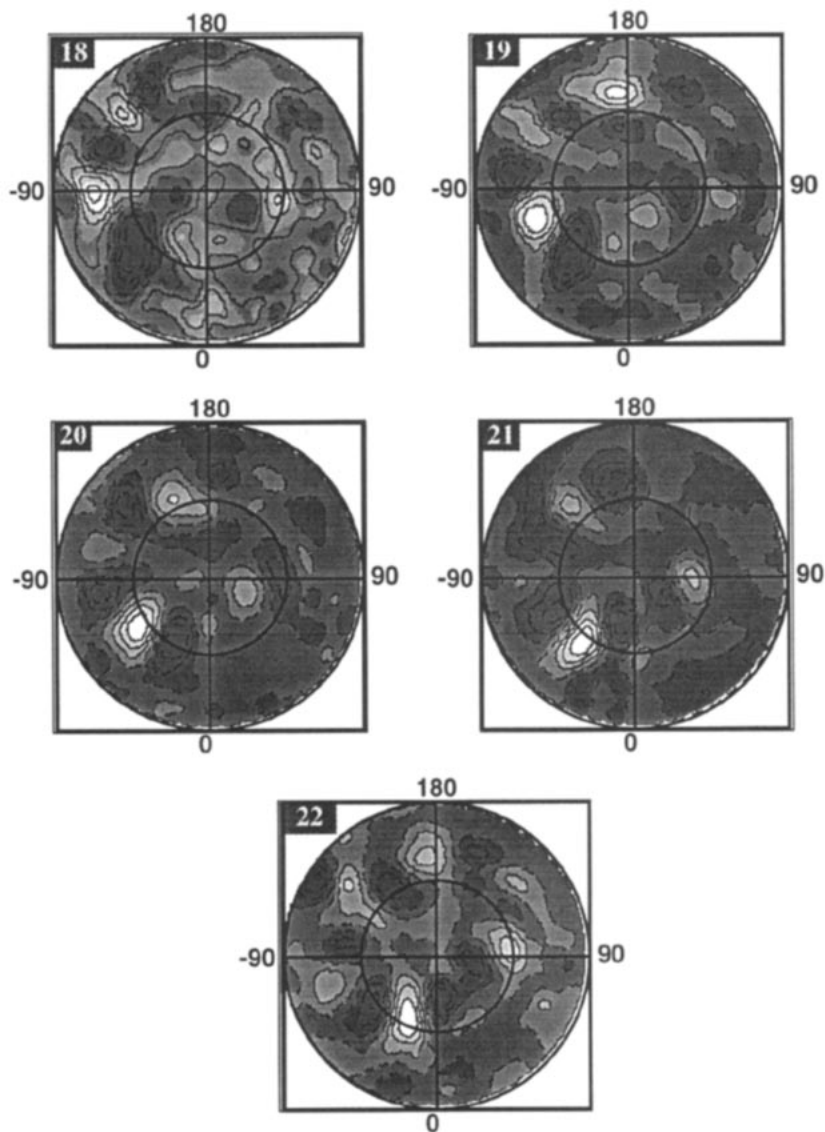


Figure 1 Five consecutive days from a January, 1989 northern hemisphere 2–6-day bandpass analysis of 200 mb geopotential. Light shading indicates high height (anticyclone) and dark shading indicates low height (cyclone); each frame is normalized to fixed maximum amplitude. The outer circle is 20°N, and the inner circle is 55°N.

- structure consists of a single high or low pressure cell. Isotropy figures importantly in nonlinear equilibration (cf Section 7).
3. The amplitude of the bandpass eddies peaks near the tropopause level, though there is typically a secondary maximum near the ground. This can be seen in the bandpass rms eddy geopotential from Lau (1978) reproduced in Figure 2. Bottom-trapped structures are not observed at any stage of the life cycles of the eddies (Lim & Wallace 1991).
 4. The synoptic eddies transport sensible heat poleward. This heat flux is bottom-trapped, with most of the flux being concentrated below 500 mb (Lau 1978, 1979). As a consequence of geostrophic balance, the poleward heat flux implies that phase lines of the geopotential tilt westward with height. Spatial spectral analysis (Hayashi & Golder 1977, Kung 1988, Randel & Held 1991) shows that the poleward heat flux peaks near zonal wavenumber 5–6, corresponding to about 4400 km.
 5. Synoptic eddy activity is most prominent in geographically localized storm tracks, which are meridionally confined to the vicinity of the midlatitudes, and—in the northern hemisphere—zonally confined within and somewhat downstream of the strong jet regions appearing over the Atlantic and Pacific oceans (Blackmon et al 1977, Lau 1978, Plumb 1986).
 6. The seasonal cycle of Pacific storm track activity shows a distinct midwinter minimum, despite the steady increase of the pole-equator temperature gradient through the winter (H. Nakamura 1992).

The synoptic eddies have a profound effect on climate. They act to even

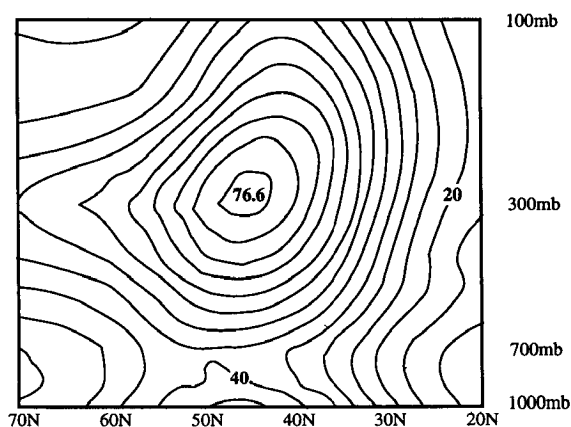


Figure 2 Latitude-height pattern of rms bandpass geopotential, from Lau (1978). Results are for winter, in the Pacific storm track sector. Contour interval is 5 m.

out the pole/equator temperature difference. They also support vorticity, momentum, and moisture fluxes. They control the extratropical vertical temperature profile which, in radiative-convective equilibrium, would otherwise relax to moist-neutral (e.g. Held 1978a, 1982; Zhou & Stone 1993b). The eddies in large measure determine the extratropical precipitation patterns. Through their influence on surface wind variability, they also affect evaporation. The resultant precipitation/evaporation balance governs the ocean surface boundary condition, which in turn affects climate through its influence on the thermohaline circulation (Broecker & Denton 1989). The eddies also govern the midlatitude cloud patterns, with important repercussions for the Earth's radiation budget (Ramanathan et al 1989). Analogous eddies (though with different time and space scales) occur on Mars (Barnes 1980, 1981; Williams 1988a,b), Jupiter (Williams, 1985, 1988a,b), and probably Saturn, where they are likely to play a similar role in climate. Synoptic eddies also arise in the Earth's oceans, in connection with instabilities of the Gulf Stream, Kuroshio, and the Antarctic Circumpolar Current.

Clearly, no meaningful treatment of climate can be undertaken without a thorough understanding of the behavior of synoptic eddies. From a theoretical standpoint, the onus for providing this understanding has fallen largely on baroclinic instability theory.

3. THE GOVERNING EQUATIONS AND THEIR ENERGETICS

Our discussion will be based on the quasi-geostrophic β -plane equations describing the evolution of a rapidly rotating, stratified fluid (see Pedlosky 1979). In the absence of thermal or mechanical damping, the potential vorticity (henceforth PV) is conserved following horizontal trajectories, i.e.

$$\partial_t q + \mathbf{v} \cdot \nabla q + \beta v = 0, \quad (3.1a)$$

where

$$q = \left(\partial_{xx} + \partial_{yy} + \frac{1}{\rho} \partial_z \frac{\rho f_0^2}{N^2} \partial_z \right) \psi, \quad (3.1b)$$

$$\mathbf{v} = (u, v) = (-\partial_y \psi, \partial_x \psi), \quad (3.1c)$$

and x is the east-west ("zonal") distance, y is the north-south ("meridional") distance, positive toward the north), and z is altitude. The Coriolis parameter is $f_0 = 2\Omega \sin(\theta_0)$ and is evaluated at the central latitude θ_0 ,

where Ω is the angular velocity of the planet; its gradient β is $(2\Omega/R)\cos(\theta_0)$, where R is the radius of the planet. The variables $\rho(z)$, $\Theta(z)$, and $N(z)$ are the background density, potential temperature, and Brunt-Väisälä frequency, respectively.

The problem is completed by specification of lower and upper boundary conditions. The lower boundary condition over a rigid surface (like that of Earth or Mars) is

$$\partial_z \psi + \mathbf{v} \cdot \nabla (\partial_z \psi + N^2 h/f_0) + \frac{N^2 \delta_e}{f_0} (\partial_{xx} + \partial_{yy}) \psi = 0 \quad \text{at } z = 0, \quad (3.1d)$$

where $h(x, y)$ is the bottom topography. The potential temperature deviation θ is related to ψ by

$$\frac{\theta}{\Theta(z)} = \frac{f_0}{g} \partial_z \psi. \quad (3.1e)$$

Bretherton (1966) has noted that (3.1d) is equivalent to putting a $\delta(z - \varepsilon)$ PV spike infinitesimally above the ground and imposing $\partial_z \psi = 0$ at the ground; hence a temperature perturbation at the ground is dynamically equivalent to a concentrated PV perturbation just above the ground. Although all of the stability results with a rigid boundary can be derived in terms of this construction, we generally opt to deal with (3.1d) explicitly instead.

The “two layer model” (see Pedlosky 1979) is a variant on (3.1) based on a highly restricted vertical structure. It is important for reasons of computational and analytic economy.

In (3.1d) boundary layer friction appears in the form of Ekman damping, characterized by an Ekman depth δ_e , defined by the relation $w_e = \delta_e (\partial_{xx} + \partial_{yy}) \psi$, where w_e is the Ekman pumping at the upper edge of the boundary layer. The Ekman depth is a product of the boundary layer model. If friction is represented by a constant Newtonian viscosity ν , then $\delta_e = \sqrt{\nu/2f_0}$; for $\nu = 4 \text{ m}^2/\text{sec}$, this yields a depth of about 140 m at midlatitudes. If friction is instead represented by a Rayleigh friction drag with damping time τ applied over a depth D , then $\delta_e = Df_0\tau/[1 + (f_0\tau)^2]$. Because of the finite depth of the friction layer in this case, δ_e never exceeds $\frac{1}{2}D$ and vanishes for both large and small τ . One can be quite confident that the friction layer does not extend very far into the interior of the atmosphere since if it did there would be a characteristic ageostrophic turning of the winds aloft, which is not observed. More realistic boundary layer models have been discussed in Lin & Pierrehumbert (1988), and it appears that values of δ_e much in excess of 200 m are not likely, particularly over the oceans. Ekman damping is the most significant form of dissipation

for baroclinic instability on the Earth. Radiative damping, which would appear in (3.1a), is probably too weak on Earth to appreciably affect the modes corresponding to synoptic eddies, though it is important on Mars, owing to its radiatively active CO₂ atmosphere.

If there is an upper boundary (as in the ocean or in laboratory experiments), another condition of the form (3.1d) is applied there. In an atmosphere, this condition is replaced by the requirement that the perturbation energy density be well behaved as $z \rightarrow \infty$, i.e. that it be decaying or bounded; if it is only bounded, then supplemental radiation conditions may be necessary, as detailed in Section 5. Gaseous planets, like Jupiter, Saturn, Uranus, and Neptune, have no rigid lower boundary in proximity to their dynamically active (and visible) outer atmospheres, and so the condition (3.1d) is replaced by a boundedness or decay condition at $z \rightarrow -\infty$.

Next, we divide the streamfunction into a background part and a perturbation part, writing

$$\psi = \psi_0(y, z) + \psi'(x, y, z, t) \quad (3.2)$$

whence the perturbation vorticity equation is

$$(\partial_t + u_0 \partial_x)q' + v'q_{0y} + \mathbf{v}' \cdot \nabla q' = 0 \quad (3.3a)$$

and the perturbation (rigid surface) boundary condition is

$$(\partial_t + u_0 \partial_x) \partial_z \psi' + v' \eta_{0y} + \frac{N^2 \delta_e}{f_0} (\partial_{xx} + \partial_{yy}) \psi' + \mathbf{v}' \cdot \nabla \partial_z \psi' = 0 \quad \text{at } z = 0, \quad (3.3b)$$

where q' , u' , and v' are defined analogously to (3.1b,c), $u_0 = -\partial_y \psi_0$, and the background state gradients are

$$q_{0y} = \beta - \partial_{yy} u_0 - \frac{1}{\rho} \partial_z \frac{\rho f_0^2}{N^2} \partial_z u_0 \quad (3.3c)$$

$$\eta_{0y} = -\partial_z u_0 + N^2 \partial_y h / f_0. \quad (3.3d)$$

We have assumed $h = h(y)$ in writing (3.3b). Using (3.1e) it can be shown that η_{0y} is proportional to the potential temperature gradient measured along the sloping lower boundary.

The perturbation energy is

$$E' = \int \frac{\rho}{2} \left[u'^2 + v'^2 + \frac{f_0^2}{N^2} (\partial_z \psi')^2 \right] dx dy dz, \quad (3.4)$$

and we wish to know when E' grows in time. The equation governing E'

is obtained by multiplying (3.3a) by ψ' , integrating over all space, and employing the boundary condition (3.3b). This equation can be written in the form

$$\frac{dE'}{dt} = \int \mathbf{Q} \cdot \nabla u_0 dy dz - \rho(0) f_0 \delta_e \int |\nabla \psi'(x, y, 0)|^2 dx dy - F_{\text{top}}, \quad (3.5a)$$

where the “ Q -vector” (Edmon et al 1980) is given by

$$\mathbf{Q} = (Q_y, Q_z) = (-\rho \overline{u'v'}, \rho \frac{f_0^2}{N^2} \overline{v' \partial_z \psi'}) \quad (3.5b)$$

in which the overbar denotes an integral over x . This vector is very convenient for diagnosing eddy properties, particularly in light of its reappearance in the wave action formulation, as will be seen shortly. The second term on the right-hand side of (3.5a) represents the energy sink due to Ekman friction, while F_{top} is a boundary term representing energy flux out of the top of the domain. The precise form of F_{top} will not concern us. It has been assumed that the analogous lateral flux terms vanish by virtue of meridional confinement in a channel, meridionally periodic boundary conditions, or rapid decay of the perturbation at large $|y|$. Note that (3.5) is true even for the fully nonlinear system.

The first integral on the rhs of (3.5a) represents the conversion of energy between background flow and perturbations. In the integrand, $Q_z \partial_z u_0$ is the baroclinic conversion term, which represents energy exchange with the potential energy of the basic state. By thermal wind balance, positive vertical shear implies colder air toward the pole, and in this situation, perturbations with a poleward heat flux tap the baroclinic energy and grow. The baroclinic term acts to release potential energy by reducing the tilt of the mean isentropes. Angular momentum conservation prevents energy release by symmetric (x -independent) overturning except where $f_0(q_0 + f_0) < 0$, a situation known as *symmetric instability* (Hoskins 1974); in the quasi-geostrophic system, symmetric motions of the required type are precluded by the constraint of geostrophic balance. When $f_0(q_0 + f_0) > 0$, baroclinic instability requires three-dimensional trajectories, and this is the situation we emphasize.

The term $Q_y \partial_y u_0$ is the barotropic conversion term, representing exchanges with the kinetic energy of the flow. In the terrestrial atmosphere it typically acts as an energy sink (Peixoto & Oort 1992, inter alia). Synoptic eddies thus act as an intermediary, transforming potential energy due to differential heating of the planet into kinetic energy of barotropic jets.

For any basic state flow with vertical or horizontal shear, an initial perturbation can be crafted which will extract energy and begin to grow.

However, as time goes on the structure of the perturbation will generally change, and growth may quickly cease or even turn to decay. The main object of stability theory is to identify energy-extracting structures which can persist long enough to amplify by a significant amount.

4. STABILITY CRITERIA

In this section we discuss the circumstances under which eddies of initially small amplitude can experience sustained growth (according to some suitable measure of eddy amplitude). This is a matter that can be most elegantly discussed with reference to the wave action budget of the perturbation. We linearize (3.3a), multiply by q' , and average over x to obtain

$$\partial_t A + \nabla \cdot \mathbf{Q} = 0, \quad (4.1a)$$

where the *wave action* (or more properly *pseudomomentum*) A is

$$A = \frac{\rho}{2} \frac{\overline{q'^2}}{q_{0y}}. \quad (4.1b)$$

An analogous budget for the variance of temperature at the ground can be obtained by multiplying (3.3b) by $\rho \partial_z \psi'$ and averaging in x . Ekman friction introduces a term of indefinite sign into this budget, in the presence of which no useful stability criterion can be obtained. Therefore, we restrict attention to $\delta_e = 0$, in which case

$$\partial_t B + Q_z(0) = 0 \quad \text{at } z = 0, \quad (4.2a)$$

where

$$B = \frac{\rho(0)}{2} \frac{f_0^2}{N^2} \frac{\overline{(\partial_z \psi')^2}}{\eta_{0y}}. \quad (4.2b)$$

Both A and B can be written in terms of meridional particle displacements ζ , since, e.g. $q' = \zeta q_{0y}$.

Integrating (4.1a) over space, assuming boundary conditions that make the horizontal flux terms vanish, and eliminating the bottom flux contribution using (4.2) yields the following relation for total pseudomomentum Π :

$$\frac{d}{dt} \Pi = - \int Q_z(y, \infty) dy, \quad \Pi = \int A dy dz + \int B dy. \quad (4.3)$$

The upper boundary integral represents pseudomomentum loss due to

radiation out of the top of the domain. When the radiation term is zero, we obtain a conservation relation which yields the desired stability criterion. Specifically, if 1. q_{0y} and η_{0y} each have uniform sign throughout the domain and also have the same sign as each other, and 2. the magnitudes of q_{0y} and η_{0y} have a nonzero lower bound and a finite upper bound, then the system is *stable* in the sense that the perturbation enstrophy

$$\mathcal{E}' = \int \frac{1}{2} \rho q'^2 dx dy dz \quad (4.4)$$

remains small for all times if it is initially small. Early treatments of the stability criterion were limited to normal modes; a limited form of the criterion was proved by Charney & Stern (1962), with the important effects of the lower boundary incorporated by Pedlosky (1964a,b). When $\eta_{0y} = 0$, the criterion states that instability is possible only if q_{0y} changes sign somewhere within the domain.

Our proof is based on the linearized form of the perturbation equations, but it is profound and important that essentially the same criterion guarantees stability under the fully nonlinear dynamics. This is a nontrivial result, as the neglected nonlinear terms could, in principal, accumulate over a long time and allow the perturbation to grow in circumstances where linear theory indicates stability. A complete discussion of the nonlinear stability criterion, based on methods pioneered by Arnol'd (1965), can be found in McIntyre & Shepherd (1987) and Shepherd (1990). The stability criterion stated above is commonly referred to as the “Charney-Stern” criterion, though with more justice it could be called the “Charney-Stern-Pedlosky” (CSP) criterion. In the general form in which we have stated it, and particularly with nonlinear extensions, one could easily justify adding on a few other initials as well.

Even in circumstances that formally preclude sustained growth of the perturbation, there is considerable scope for significant amplification. First, the perturbation enstrophy can grow by a factor $[\max(q_{0y})]/[\min(q_{0y})]$ if the perturbation PV is initially localized in a region of small q_{0y} and later moves to a region of large q_{0y} ; an analogous argument can be made with respect to surface temperature fluctuations and η_{0y} , or with regard to conversions between the A term and B term in (4.3). Second, perturbation energy can grow even if perturbation enstrophy is not growing. This is because the energy and enstrophy norm are not *equivalent norms*—a sequence that converges to zero in energy need not converge to zero in enstrophy, because of the higher order derivatives involved in the latter. The ratio of energy to enstrophy defines a length scale $(L_P)^2 = E'/\mathcal{E}'$, which is a function of time if the structure of the perturbation is evolving.

Energy growth for fixed enstrophy requires growth of L_p . The initial perturbation can be made to have arbitrarily small L_p by giving it a very oscillatory structure in any direction. If the structure “unscrambles” and evolves to a smoother state, L_p increases and energy grows. L_p may be bounded above, depending on the geometry and the assumptions about perturbation behavior at $z = \infty$; if it is, the resulting bound on E' is known as *Poincaré's inequality*. [See the discussion of Arnol'd's second theorem in Shepherd (1990).] In the absence of arguments bounding L_p below, however, the inequality still does not limit possible energy growth.

The archetypical scenario permitting instability in the terrestrial atmosphere involves $q_{0y} \geq 0$ everywhere, $\partial_z u_0 > 0$ at the ground, and a sufficiently flat lower boundary to keep η_{0y} negative. This condition is almost ubiquitously satisfied throughout the midlatitude troposphere in both hemispheres. In this situation, growth of the PV perturbation aloft is accompanied by simultaneous growth of the surface temperature perturbation. A growing disturbance can be regarded as a coupled PV/surface-temperature motion, with the PV anomaly inducing a motion that stirs up the surface temperature and the surface temperature in turn stirring up the PV aloft. Equation (3.1b) implies that the influence depth of a PV or surface temperature perturbation of length L is the “deformation depth,” $f_0 L/N$, provided this is not much in excess of the density scale height H . If the PV disturbance is at altitude D , then motions with $L \ll ND/f_0$ will have the PV dynamics decoupled from the surface dynamics, and cannot be expected to cohere and grow. This implies that shortwave instabilities must also be shallow, and that deep modes filling out a density scale height must have horizontal scales comparable to the radius of deformation $L_d = NH/f_0$, if not longer. These arguments are quite general, and carry over to strongly nonlinear motions.

Sufficiently extreme vertical structure in N or in u_0 can make q_{0y} change sign internally and allow instabilities that do not rely on interaction with the rigid boundary. This is particularly important to the prospect for baroclinic instability on the gas giant planets. A positive jump in N^2 , such as happens at a tropopause, produces a negative δ -function in q_{0y} if the vertical shear at the tropopause is *negative*, and this permits instability if $q_{0y} > 0$ elsewhere. In this case, the PV disturbance at the tropopause plays a role analogous to the surface temperature disturbance at a rigid boundary. The apparent singularity in (4.1b) disappears if A is written in terms of meridional particle displacements.

Inviscid stability criteria can also be derived from other conservation laws (Shepherd 1990) but typically apply in much more specialized circumstances than the pseudomomentum criterion.

5. NORMAL MODES

We are free to seek solutions to (3.3) of the special form

$$\psi'(x, y, z, t) = \Psi(y, z) e^{ik(x-ct)}, \quad (5.1)$$

where k is the wavenumber in the x (zonal) direction and c is the complex phase speed; the complex frequency is $\omega = kc$. Substituting in (3.3) and dropping the nonlinear terms results in the equation

$$(u_0 - c) \left(\frac{1}{\rho} \partial_z \frac{\rho f_0^2}{N^2} \partial_z + \partial_{yy} - k^2 \right) \Psi + q_{0y} \Psi = 0 \quad (5.2)$$

with lower boundary condition (for a rigid lower surface)

$$(u_0 - c) [\partial_z \Psi] + \eta_{0y} \Psi - i \frac{N^2 \delta_c}{f_0 k} (-k^2 + \partial_{yy}) \Psi = 0 \quad \text{at } z = 0. \quad (5.3)$$

At the upper boundary, we impose an energy-density decay condition or a radiation condition. If N or the shear have discontinuities, as is common at a tropopause, then singularities appear in the coefficients of (5.2). These must be handled through analytically derived jump conditions. Integrating (5.2) across a line of singularity yields the condition

$$(u_0 - c) \left[\frac{\partial_z \Psi}{N^2} \right]_{z_c - \epsilon}^{z_c + \epsilon} - \Psi[y, z_c(y)] \left[\frac{\partial_z u_0}{N^2} \right]_{z_c - \epsilon}^{z_c + \epsilon} = 0, \quad (5.4)$$

in which we have also allowed for a jump in the vertical shear. Comparison of (5.4) with (5.3) shows that there is (apart from the Ekman term) a dynamical equivalence between the jump condition for profiles with discontinuous shear and a rigid lower boundary condition.

The linear, homogeneous system (5.2, 5.3) defines an eigenvalue problem for c . Generally, it must be solved numerically. The most straightforward method is to discretize the operator using spectral or finite-difference methods, and solve the resulting matrix eigenvalue problem using a global eigenvalue iteration (Lin & Pierrehumbert 1988 and references therein). The drawback of this method is the large amount of work required for two-dimensional (2-D) jets $u_0(y, z)$. With m degrees of freedom in each direction, the memory requirement grows like m^4 and the computation time grows like m^6 . It is essential to use a stretched vertical coordinate, because high resolution is needed near the ground, and though less resolution is needed aloft, the upper boundary must be removed to two or more scale heights to prevent spurious lid effects. Pressure coordinates take care of this nicely, and with the hydrostatic relation the required

transformation is simply $\partial_z = -\rho g \partial_p$. With suitably chosen pressure coordinates and second-order vertical differences, 20 gridpoints in the vertical typically suffice to adequately resolve the most unstable mode. This resolution is somewhat better than the 9–15 levels of vertical resolution employed in most tropospheric general circulation models; it is interesting that such models reproduce observed synoptic eddy features despite their marginal resolution (see, e.g. Lau & Nath 1987). Two-dimensional stability computations up to $m = 20$ are accessible on many workstations. For one-dimensional (1-D) problems $u_0(z)$, the required effort is much less because the y -dependence can be assumed sinusoidal; the global eigenvalue approach is then clearly the method of choice.

Various alternate strategies involve a Newton-Raphson iteration on c to zero out the determinant of the system (5.2) (e.g. Lin & Pierrehumbert 1993). These can be very efficient, because the determinant can be evaluated quickly and uses very little storage. With second-order vertical differences, for example, (5.2) becomes block-tridiagonal which leads to considerable computational economies. In one dimension, this reduces to the shooting method familiar from ordinary differential equation problems. A considerable drawback is that convergence is problematic without a good initial guess for c ; further, it is difficult to ensure that one has found all the unstable modes. A good hybrid strategy in two dimensions is to use the iteration methods to refine the accuracy of estimates obtained from the global eigenvalue approach.

A third approach is to replace c in (5.2) by $-k^{-1}\partial_t$, and solve the resulting 2-D initial value problem. After sufficiently long integration, the most unstable mode will naturally emerge. This method is storage-efficient but convergence can be slow, and it is cumbersome (though possible through re-orthogonalization) to obtain anything other than the most unstable mode for each k .

Regardless of the method employed, the jump condition (5.4) must be respected. If second-order finite differences are used to represent the vertical operator in the flux form appearing in (5.2), the jump condition will be taken care of automatically, provided that the same difference is used to evaluate the vertical derivatives appearing in $\partial_z q_0$.

Basic Behavior of the Eigenvalues and Eigenmodes

The Charney problem is defined by the profile $u_0 = az$, $N = \text{const.}$, $\rho = \rho_0 e^{-z/H}$ (Charney 1947), with $\delta_e = 0$. Certain features of the Charney problem repay close study, because they are generic to the stability of virtually all profiles with an appreciable horizontal temperature gradient at the ground. We may set $\Psi = \Phi(z) \exp(i ly)$, where l is the meridional wavenumber. If we nondimensionalize (5.2) by choosing the radius of

deformation $L_d = NH/f_0$ as the unit of horizontal length, H as the unit of depth, and aH as the unit of velocity (whence the unit of time is N/af_0), the eigenvalue problem for the nondimensional phase speed is controlled by only two parameters: the nondimensional wavenumber $\kappa^2 = (k^2 + l^2)L_d^2$ and the Charney number $\beta^* = \beta L_d^2/(aH)$.

For typical tropospheric conditions, $\beta^* < 1$. When β^* is order unity or less, the wavelength of the most unstable mode scales with the radius of deformation, and the modes fill out the scale height H . However, when $\beta^* \gg 1$, the Charney number can be absorbed into the vertical coordinate, yielding a new depth scale $(f_0/N)a/\beta$ and a new length scale a/β . The characteristic frequency is still (af_0/N) . Thus, in the limit of weak shear or large N , the most unstable mode becomes shallow, its wavelength becomes short, and its growth rate approaches zero (Held 1978b).

The problem was formulated by Charney (1947), who also obtained some results on the behavior near the long-wave stability boundary. The full dispersion relation of the Charney model was first obtained by Kuo (1952, 1953), by means of truly prodigious feats of evaluation of hypergeometric functions on a mechanical desktop calculator. Miles (1964) noted that there is a simple physically illuminating solution to the problem in the short-wave limit, which shows that all (undamped) flows with nonvanishing vertical shear at the ground are unstable. This approximate solution in the absence of Ekman friction is (Miles 1964, McIntyre 1972, Pierrehumbert 1986):

$$\omega = u_0(0)k + \frac{f_0}{N} \partial_z u_0(0) \left(\frac{k}{|k|} + \left\{ \left[\frac{1}{2} + (1 + \beta^*)p \right] + i \frac{\pi}{e^2} (1 + \beta^*) \right\} (kL_d)^{-1} \right), \quad (5.5)$$

where $p \cong -0.6705$. Formula (5.5) is valid for $\partial_z u_0(0) > 0$; it is written for $l = 0$, but can be generalized to finite l through rotation of the (x, y) coordinate system. The corresponding lowest order modal structure is $\Phi(z) = \exp[(-kNz)/f_0]$. Because the solution is bottom-trapped, it is universal to all profiles with vanishing horizontal shear at the ground. From a PV standpoint, the short waves can grow because the mode becomes shallow, and the PV anomaly stays close enough to the lower boundary to remain coupled to the surface dynamics. The most unstable mode for flows with appreciable shear at the ground is generally a continuation of this short-wave branch, and is similar in character. Modes of this type are known as “Charney Modes,” as they are in essence the ones identified in his original paper.

Figure 3 shows some numerically computed dispersion relations. Growth rate and phase speed are given as a function of zonal wavenumber

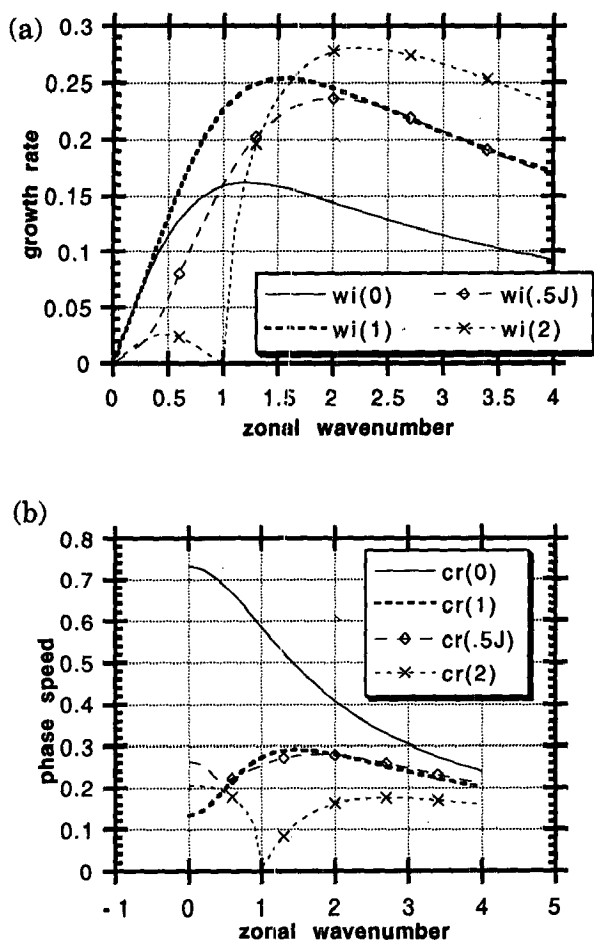


Figure 3 (a) $\omega_i(k)$ and (b) phase speed c_r for the Charney model $\beta^* = 0, 1, 2$, and for a jet flow with tropopause and $\beta^* = 0.5$ based on shear at the ground. Curves are computed with meridional wavenumber $l = 1$, corresponding to a channel of half width π radii of deformation.

k , for fixed meridional wavenumber $l = 1$; this corresponds to flow confined in a midlatitude channel. The most unstable modes have wavelengths ranging from about πL_d (roughly 2000 km) to about $2\pi L_d$, and are consonant with the observed synoptic scales. In accordance with (5.5) the growth rate has a gradually decaying short-wave tail, so that the inviscid instability has a broadband character.

As suggested by scaling considerations, increasing β^* pushes the instability to shorter wavelengths. As β^* is increased from zero to unity, the

nondimensional growth rate increases markedly, showing that increasing the interior PV gradient helps the disturbance to extract energy more rapidly. Further increases in β^* yield relatively little increase in the nondimensional growth rate. We also show results for the jet profile $u_0 = z \exp[-z^2/(2D^2)]$, with a tropopause at $z = D$ characterized by a doubling of N^2 ; we chose $D = 1.2H$ and $\beta^* = 0.5$ (based on the shear at $z = 0$). The results differ little from the $\beta^* = 1$ Charney profile case, which has similar mean tropospheric shear. The asymptotic form (5.5) based on the larger ground level shear does not take hold until considerably larger wavenumbers.

Except for $\beta^* = 0$, the phase speeds are all indicative of a lower-tropospheric steering level and are thus compatible with the observed synoptic eddies. Even for $\beta^* = 0$, the steering level approaches the upper troposphere only at long waves. For very weak shear ($\beta^* = 2$), the Green/Burger/Miles modes (cf Pedlosky 1979) can be seen for $k < 1$. These long-wave solutions have their energy source in the troposphere, but radiate deeply into the stratosphere; in consequence, unlike the Charney modes, they are highly sensitive to wind conditions aloft (Gall 1976a).

Figure 4 shows the vertical structure of the eigenmodes at $kL_d = 1$ and 2, for the internal jet case. For the short-wave mode, the geopotential is bottom-trapped, exhibiting a vertical structure very unlike the observations. The shallow heat-flux, on the other hand, is rather like the observations. Likewise, the 60° phase shift between the ground and the lower troposphere is consistent with observed bandpass eddy phase shifts (Lau 1979). The PV perturbation is sharply peaked at the steering level. Its phase near the steering level is almost coincident with that of the surface temperature perturbation, and tilts sharply westward with height. For the longer wave case, $kL_d = 1$, the net phase shifts are not too different from $kL_d = 2$, though they are spread over a deeper layer and perturbation quantities are less vertically trapped generally. In this case, the geopotential perturbation has a secondary maximum aloft, and is not so dramatically at odds with the observations as was the case for $k = 2$. In Sections 6 and 7 we discuss some reasons why the longer waves may come to dominate the bandpass statistics.

The energy budget (3.5) shows that Ekman damping acts as a strong energy sink for bottom-trapped modes; hence, it will greatly affect the short-wave behavior of the instability, and perhaps also the most unstable mode. With the nondimensionalization employed above, the Ekman term in the nondimensional boundary condition appears with the coefficient δ_c/Δ , where

$$\Delta = (kL_d) \left(\frac{a}{N} \right) H. \quad (5.6)$$

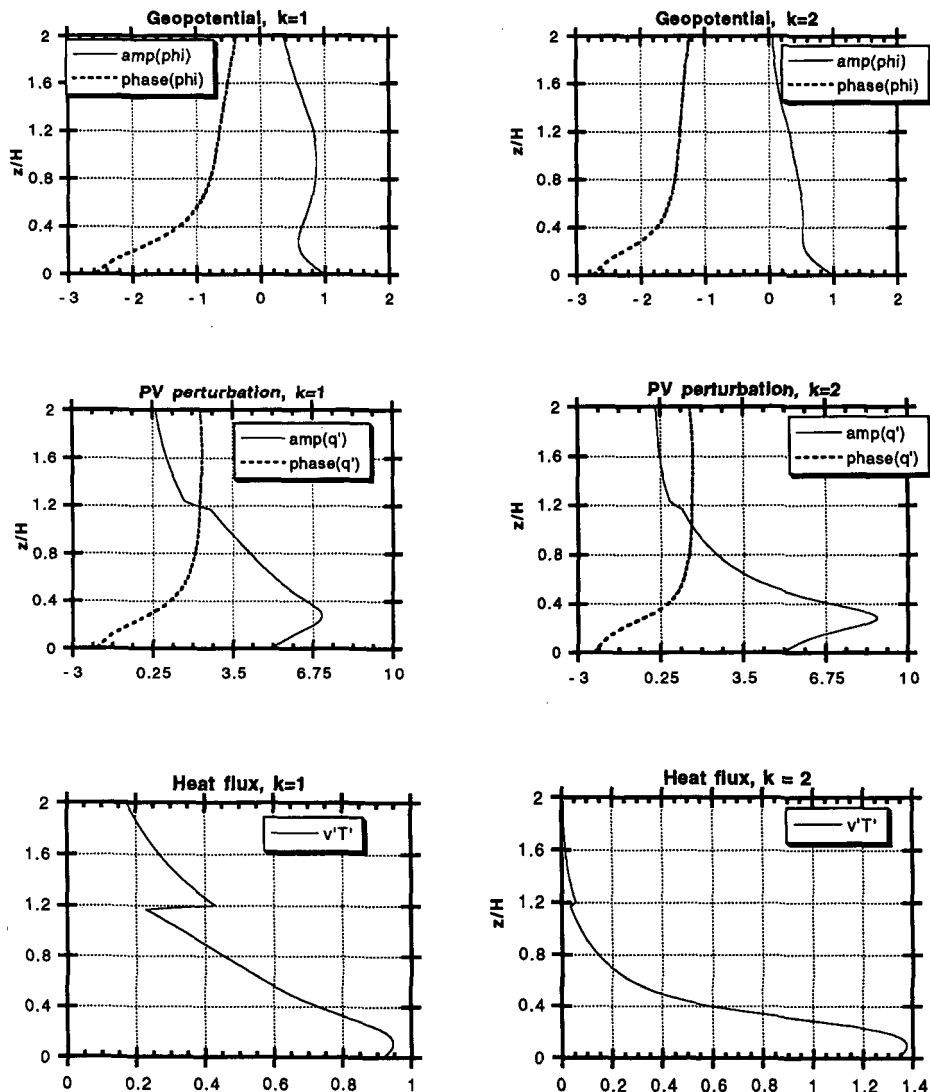


Figure 4 Eigenmode structure for 1-D jet flow. We show the amplitude and phase of geopotential perturbation, amplitude and phase of the potential vorticity perturbation, and the heat flux $v'T'$. Phases are given relative to the phase of the ground-level temperature perturbation. Left panels: $k = 1$. Right panels: $k = 2$.

This implies that for fixed kL_d , the Ekman damping becomes less important as shear is increased or N is decreased. Ekman friction introduces a short-wave cutoff, reduces the growth rate of the most unstable mode, and can completely stabilize flows with weak shear (Card & Barcilon 1982, Lin & Pierrehumbert 1988). It doesn't appear likely that Ekman damping can completely stabilize the system for earthlike parameters (Valdes & Hoskins 1988, Lin & Pierrehumbert 1988), though a 50% reduction in growth rate is possible over the extensive northern hemisphere continents. Ekman stabilization is quite inconsequential in the oceanic storm track regions (Lin & Pierrehumbert 1993). Though Ekman friction increases scale selectivity, nonlinearity may be a bigger part of the scale selection process, as we shall see later.

Now let's consider some dimensional values pertinent to the internal jet case of Figure 3, but with $\beta^* = 0.15$. This case has a maximum non-dimensional growth rate of 0.22, and phase speed 0.33 occurring at $k = 1.9$. With $H = 7$ km, $f_0 = 10^{-4} \text{ sec}^{-1}$, and $N = 0.01 \text{ sec}^{-1}$, we have $L_d = 700$ km and $\partial_z u_0(0)$ approximately $4 \times 10^{-3} \text{ sec}^{-1}$. This corresponds to a modest jet maximum of 20 m/s at $z = 1.2H$. The dimensional growth rate is about 0.75 day^{-1} and the phase speed is 9.25 m/s, both of which are consistent with observations. Increasing the jet strength reduces β^* , which doesn't much alter the nondimensional growth rate; however, the dimensional growth rate would increase considerably. Reducing N with shear fixed reduces β^* and shifts the instability to shorter dimensional wavelengths but also increases the growth rate, in accordance with the characteristic rate $(f_0/N)\partial_z u_0(0)$. These short-wave disturbances are nonetheless deep, troposphere filling modes, because the tropospheric radius of deformation is also short.

The scaling with N leads to the premise that baroclinic instability would produce a statically stable extratropical atmosphere on a rotating, differentially heated planet even in the absence of moisture (cf Held 1978a, 1982). Local radiative-convective equilibrium would drive N to zero but would have a strong pole-equator temperature gradient. Such a flow would be violently unstable to baroclinic instability. In quasi-geostrophic theory, the growth rate diverges as $N \rightarrow 0$, but ageostrophic effects limit the divergence for sufficiently small N (Stone 1966, 1970; Nakamura 1988). It is presumed that instability increases N so as to adjust toward a less unstable state. Eddies could do this by taking cold air manufactured in polar regions and sliding it under warmer air, and vice versa for warm air produced near the tropics.

The above picture is based on the 1-D problem confined in a channel. More elaborate calculations with realistic 2-D jet profiles $u_0(y, z)$ offer no surprises in the case when the jet is symmetrical about a peak at $y = y_m$.

In this case, the main effect of the meridional shear is to provide the confinement that must be imposed by artificial channel walls in the 1-D problem (Lin & Pierrehumbert 1983, Iouannou & Lindzen 1986); it also allows the linear modes to have some momentum flux. The stability problem for realistic 2-D jets has been discussed by Gall (1976a) and Simmons & Hoskins (1976), among others.

The chief novelty in two-dimensional jet calculations arises in situations with very asymmetric barotropic flow. In the extreme, we may consider $u_0(y, z) = G(z) + by$, with $G(z)$ chosen to permit vigorous baroclinic instability when $b = 0$. A variant of this problem was addressed by James (1987). Increasing b doesn't alter the PV gradients or the potential energy source. Yet, calculations show that the growth rate goes to zero with increasing b , due to rapid horizontal shearing out of the eddies, which gives them a shape implying a strong barotropic energy sink. The eddies are literally torn apart before they can grow much. Nonlinear evolution spontaneously generates barotropic shears, which can then help shut off growth. This is known as the "barotropic governor" effect (James 1987, Nakamura 1993a,b).

Vertical Trapping of Unstable Modes

There is an intimate connection between vertical trapping, vertical Rossby wave propagation, and amplification. Suppose that u_0 and N are both constant at $z \rightarrow \infty$. The large- z solution can be decomposed into meridional modes with wavenumber l , and the behavior for each l is

$$\Phi(z) = \rho^{-1/2}(\mathbf{I} e^{-imz} + \mathbf{R} e^{imz}), \quad (5.7)$$

where m is determined as a function of ω by inverting the Rossby wave dispersion relation $\omega(k, m)$. Let c_r and c_i be the real and imaginary parts of the phase speed. For $\omega_i = 0$, the waves are evanescent ($m^2 < 0$) for sufficiently large $u_0 - c_r$, and short waves most easily and strongly trapped (Charney & Drazin 1961). It is easily shown that nonzero c_i enhances the trapping in evanescent cases. The real part of the phase speed c_r is kept rather small for the growing modes by the requirement that the PV perturbation couple strongly to the lower boundary, which is one reason that strongly growing modes tend to be vertically trapped.

Even when $u_0 - c_r$ is small enough to permit vertical propagation of Rossby waves, temporal amplification causes vertical evanescence (Charney & Pedlosky 1963). The physical mechanism is clearest in the limit of small ω_i , in which case a Taylor series expansion of the dispersion relation implies $m_i = \omega_i/c_{gz}(m_r)$, where $c_{gz} = d\omega/dm$ is the vertical group velocity of the Rossby wave. The low-level instability may be regarded as supplying a temporally amplifying forcing to the bottom of the wave propagation

region, and vertical decay is simply due to the finite propagation time of the Rossby wave. We thus expect that when amplification in the low-level source region shuts off in the course of nonlinear equilibrium, a wave front propagates upward, below which the wave intensity becomes independent of height.

The preceding limit also supplies the desired radiation condition for modes that become untrapped as $c_i \rightarrow 0$. For arbitrary real m we have

$$Q_z = \frac{km}{2} \frac{f_0^2}{N^2} [|\mathbf{R}|^2 - |\mathbf{I}|^2]. \quad (5.8)$$

Adopting the convention $k > 0$, we can show that the term proportional to $\mathbf{I}$ in (5.7) continues into a vertically divergent solution for $c_i > 0$. Hence, we must have $\mathbf{I} = 0$ if the solution is to be the neutral limit of an unstable mode, whence $Q_z > 0$.

The above results can be extended to the case of gradually varying $u_0(z)$ by WKB methods. See Lindzen & Rosenthal (1981) for a general treatment of baroclinic instability along these lines.

Stability Boundaries for Inviscid Flow

Let λ be a parameter controlling the stability problem, so $c = c(\lambda)$; λ could be the zonal wavenumber k , but other choices are possible. A *stability boundary* is a point λ_c such that $c_i(\lambda_c) = 0$, but $c_i(\lambda) > 0$ to the right or left of λ_c (or both). Stability boundaries are of interest because they serve as anchor points for iterative numerical methods, and because they can provide some insight as to the physical character of the instabilities continuing from the neutral boundary mode. For inviscid flow the coefficients of (5.2, 5.3) are real and so both c and its complex conjugate c^* are solutions. Hence, stability boundaries involve coalescence of a conjugate pair rather than a simple zero crossing, and therefore require some form of nonanalytic behavior. For finite dimensional matrix eigenvalue problems, this generically takes the form of coalescence at a square-root branch point, with a pair of real c on one side and a complex conjugate pair on the other. Since the dispersion relation for baroclinic instability is governed by a differential equation, more exotic behavior is possible, notably logarithmic branch points associated with the singularity of (5.2) at points where $(u_0 - c) = 0$. The behavior at an inviscid stability boundary is highly constrained by the action conservation laws. The following exposition is an updated and somewhat generalized version of arguments originating in unstratified shear flow which have worked their way into the geophysical fluid dynamics literature in diverse forms (Bretherton 1966, Lindzen et al 1980 and references therein).

At the stability boundary, the time derivatives in (4.1) and (4.2) vanish. Hence $\nabla \cdot \mathbf{Q} = 0$ in the interior and $Q_z(y, 0) = 0$ wherever the action density is defined. The eigenmode equations are singular at critical lines, where $u_0 = c$. On these lines we must get $\nabla \cdot \mathbf{Q}$ through a limiting process on c .

To get the general jump condition, divide (5.2) by $(u_0 - c)$, multiply by $-ik\Psi^*$, and take the real part. At the stability boundary, we have $c = c_r + i\varepsilon$, with $\varepsilon > 0$, so by Cauchy's formula

$$\frac{1}{u_0 - c} = \frac{1}{|\nabla u_0|} \left[P\left(\frac{1}{\zeta}\right) + i\pi\delta(\zeta) \right] \quad (5.9)$$

in the vicinity of the critical layer, where ζ is the distance normal to the line $u_0 = c_r$ and P denotes Cauchy principal value. Carrying out the calculation yields

$$\nabla \cdot \mathbf{Q} = -\frac{1}{2}k\rho \left[\frac{\pi q_{0y} |\Psi|^2}{|\nabla u_0|} \right] \delta(\zeta) \quad (5.10)$$

in which the quantity in brackets is to be evaluated at the critical line.

We also need a boundary condition on Q_z at the ground. If the critical line doesn't intersect the ground, then (4.2) implies $Q_z(y, 0) = 0$ at the stability boundary. If the critical line intersects the ground, Q_z can be nonzero only at the points of intersection. If the critical line is *coincident* with the ground, then $Q_z(y, 0) = -\frac{1}{2}\pi k\rho(0) (f_0^2/N^2) (\eta_{0y}/|u_{0z}|) |\Psi(y, 0)|^2$.

In the special case in which $u_0 = u_0(z)$, then $\zeta = z$ and (5.10) can be integrated across the critical line to yield the jump condition:

$$Q_z(z_c + \varepsilon) - Q_z(z_c - \varepsilon) = -\frac{1}{2}k\rho \left[\frac{\pi q_{0y} |\Psi|^2}{|du_0/dz|} \right]_{z=z_c}. \quad (5.11)$$

In consequence of this and the boundary condition, there are only four possibilities for monotonic u_0 : (a) The critical line moves to a point where $q_{0y} = 0$. It can be shown by a straightforward perturbation theory on λ that such modes always continue into unstable modes on one side of λ_c . (b) c_r takes on a value permitting upward radiation at $z = \infty$, $Q_z(\infty) > 0$, and hence $q_{0y}(z_c) < 0$. This is a *radiating stability boundary*. If $q_{0y} > 0$ everywhere but $\eta_{0y} < 0$, a radiation stability boundary can also occur if the critical level moves to the bottom boundary. (c) c_r moves outside the range of u_0 , so there is no critical line. The modes must be nonradiating. This kind of stability boundary corresponds to a degeneracy between a pair of nonsingular Rossby waves, and can be interpreted as a *resonant wave stability boundary*. It is precluded by the semicircle theorem for

nonrotating unstratified flow, but the β effect in Pedlosky's modified semicircle theorem (Pedlosky 1979) permits nonsingular resonant behavior. (d) We could have $\Phi = 0$ at the critical line. This is a rare circumstance, because imposing $\Phi = 0$ adds a third boundary condition to a second-order problem, and makes it overdetermined. An exception to this argument occurs when the critical line approaches the bottom boundary, allowing a trapped mode with $c_r = u_0(0)$ but $Q_z = 0$ everywhere.

For the Charney problem (a) and (b) are precluded. The neutral points are of type (d) with $c_r = u_0(0)$, though they are also degenerate points and therefore have something of the character of (c) as well (Held et al 1985). Type (d) modes often stem from nonsingular Rossby modes with real $c < u_0(0)$, whose phase speed crosses into the range of u_0 , whereafter the modes develop a critical level and continue into instabilities.

The situation for non-monotonic u_0 is not much different from the above if $q_{0y} > 0$ everywhere. However, if q_{0y} changes sign in the domain, many new possibilities are opened up, since a wave can be radiated at one critical level and absorbed at another. These situations, like the radiating stability boundary, involve singular behavior of q' at the stability boundary and are therefore difficult to treat accurately by numerical means alone. For 1-D problems, it is possible to integrate around the singularity in the complex z -plane (Boyd 1985), but analogous methods for 2-D jets $u_0(y, z)$ have not yet been developed.

Stability boundaries for 2-D jets have not been extensively discussed, but they are still greatly constrained. Equation (5.10) shows that Q acts like the velocity of an incompressible fluid with source concentrated on the critical line (and at the ground). This property restricts the disposition of the line. For example, if q_{0y} has like sign everywhere, it is impossible for a stability boundary to occur at a critical line $u_0 = c_r$ which takes the form of a closed curve.

Ekman friction does not alter (5.10), but alters the boundary condition to require $Q_z(0)$ to be nonzero at the stability boundary. This, in turn, requires the critical line to be located at a point of nonzero q_{0y} . Such modes have a logarithmic singularity at the critical line, and require special treatment as for singular modes arising from non-monotonic u_0 . If the mode has nonzero amplitude at the ground and barotropic shear is not an energy source, the energy budget (3.5) requires that the Ekman-stabilized neutral modes carry a net heat flux. The potential energy released by this heat flux is balanced by energy dissipation at the ground.

The stability boundary at $k = 0$ is a special case, since $\omega_i \rightarrow 0$ with c_i finite. As a consequence of the semicircle theorem (Pedlosky 1979), c_i cannot diverge as $k \rightarrow 0$. Typically c_i approaches a nonzero constant, so $\omega_i \rightarrow 0$ linearly as $k \rightarrow 0$.

Continuum Modes

For completeness, the unstable modes must be supplemented by a spectrum of neutral modes with real c . The modes with c outside the range of u_0 (if any) are nonsingular. They can represent discrete Rossby waves (Held et al 1985), or conservative (“elastic”) scattering of Rossby waves coming in from $z = \infty$. If c lies in the range of u_0 , more interesting things can happen. Suppose that $u_0 = u_0(z)$, with u_0 and N constant at $z = \infty$, so (5.7) applies asymptotically, and let c take on a value such that m is real. For simplicity, suppose further that u_0 is monotonic, so there is only one critical line. We relax the radiation boundary condition to permit an incoming wave from $z = \infty$, so $I \neq 0$. Although the mode is no longer a stability boundary, we must still obtain it from the limiting process $c_1 \rightarrow 0^+$, for this is the realizability condition that ensures that the solution can be obtained from an initial condition consisting of a wave packet incoming from $z = \infty$ and impinging on an initially undisturbed low-level flow. Equation (5.8) together with the jump condition (5.11) then immediately implies that (with $k > 0$)

$$\frac{km}{2} \frac{f_0^2}{N^2} [|R|^2 - |I|^2] = - \left[\frac{\pi q_{0y} |\Psi|^2}{|du_0/dz|} \right]_{z=z_c}. \quad (5.12)$$

If $q_{0y}(z_c) > 0$ the incident wave is partially absorbed at the critical line. If $q_{0y}(z_c) < 0$, however, the reflected wave has higher amplitude than the incident wave, representing a novel and intriguing form of instability known as *over-reflection*. Radiating instability is an extreme case of over-reflection, since in that case no incident wave is required. Over-reflection has a long history in fluid mechanics (McIntyre & Weissman 1978 and references therein), but the first appearance of the concept may have been in early attempts at relativistic quantum theory, in which it was found that a sufficiently steep potential can over-reflect an electron beam (see the discussion of this in Pais 1991). Certain amplifying fluid disturbances can be understood in terms of modified over-reflecting waves in a cavity (e.g. Lindzen et al 1980), and there have been attempts to identify *all* shear instabilities with over-reflection. The identification can only be cleanly stated in the vicinity of a stability boundary, and not all stability boundaries lend themselves naturally to this interpretation. There is certainly an intimate connection between critical lines and instability, but we find the “mixmaster” picture put forth in Section 7 to be more physically appealing.

Finally, there can be “bound state” singular continuum modes, with $c = u_0$ somewhere, but with decaying energy density as $z \rightarrow \infty$. The essen-

tial point is that (5.2) allows q' to have a δ -function singularity where $u_0 - c = 0$. This is equivalent to introducing a jump in $\partial_z \Phi$ across the critical line. Note that these modes don't have to satisfy the previously stated jump constraint because they are not an analytic continuation of a nonsingular mode, i.e. they are not a real-axis boundary value of an analytic function. The modes at a stability boundary, in contrast, are obtained by the limiting process $c_i \rightarrow 0$. We can put the δ -function at any z_c compatible with trapping at $z \rightarrow \infty$, and so the modes make up a continuum which can be indexed by z_c . When q_{0y} is nonzero, q' has a $1/(z - z_c)$ singularity near z_c as well as the δ -function at z_c . In isolation, these modes are unphysical, but continuous superpositions of them yield a smooth field. The appearance of singular modes in the eigenvalue problem reflects the capability of the inviscid initial value problem to generate arbitrarily small scale from smooth initial data as $t \rightarrow \infty$.

The general principles in Held (1985) show that the neutral modes are orthogonal with respect to an inner product based on the pseudomomentum (4.3). When the CSP stability criterion is violated, neutral modes can have zero or even negative pseudomomentum, which leads to some novel and counterintuitive behavior of projections, superpositions, and alterations of the modes under perturbation of the system.

There are no *unstable* continuum modes, because the coefficients of (5.2), regarded as functions of c , have no singularities off the real c axis. Because of destructive interference, the energy of continuous superpositions of the singular neutral modes decays algebraically at large times. However, as will be seen shortly, the energy can exhibit appreciable—in fact unbounded—growth before its asymptotic decay sets in.

6. SPATIO-TEMPORAL DEVELOPMENT

Real perturbations arise from spatially localized excitation, and the resulting initial value problem can profitably be examined in two extremes. In the first extreme, a superposition of normal modes with various k and l is used to build up a perturbation that is initially localized in the horizontal; the vertical structure of such perturbations is constrained by the normal-mode structure. In the second extreme, the wave is assumed sinusoidal in x , but the initial condition is allowed to have an arbitrary vertical structure.

Linear Wave Packet Evolution

Consider a domain unbounded in x . We may superpose normal modes $\Psi(y, z|k)$ so as to obtain an initial condition that is zonally localized. For the limit of a $\delta(x)$ initial condition, the subsequent evolution is

$$\psi'(x, y, z, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \Psi(y, z|k) e^{i[(x/t)k - \omega(k)]t} dk. \quad (6.1)$$

We are interested in the asymptotic behavior of this integral as $t \rightarrow \infty$ with x/t fixed. With real ω , this can be solved by the familiar stationary phase method. When ω is complex, a version of stationary phase in the complex plane can be used, and the dominant contributions come from the saddle points

$$\frac{d\omega}{dk}(k_s) = \frac{x}{t}. \quad (6.2)$$

These saddle points are generally off the real k -axis, and considerable subtleties are introduced by the necessity of going into the complex plane. Not all the saddle points contribute to the asymptotics; a precise formulation of the asymptotics was given in Briggs (1964), and an algorithm for implementing the calculation has been given in Pierrehumbert (1986). The early work was motivated by problems in plasma physics, but interest in other fluid dynamical applications arose in a number of fields in the 1980s. A considerable body of literature has accumulated, and Huerre & Monkiewicz (1990) review the current state of the subject. The structure of amplifying baroclinic wave packets was first discussed by Merkin (1977), Simmons & Hoskins (1979), and Merkin & Shafranek (1980).

The quantity of primary interest is $\omega_i[k_s(x/t)]$, which is the growth rate seen by an observer moving with speed (x/t) . A calculation for the inviscid Charney model with $\beta^* = 0.5$ and $l = 1$ is shown in Figure 5. The calculation was done with $u_0(0) = 0$, but results for other surface winds can be obtained by uniform translation of all velocities. There are three

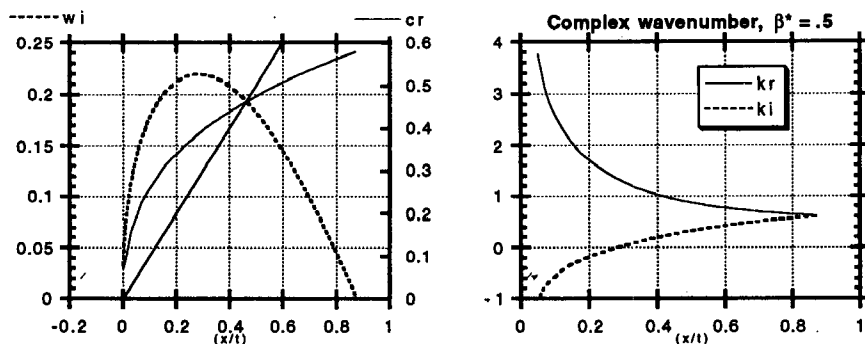


Figure 5 Growth rate, phase speed, and complex wavenumber as a function of position x/t within the wave packet for the spatio-temporal Charney problem (see text for details). The bold straight line in the left panel is $c_r = x/t$.

distinguished rays. The value of x/t for which ω_i takes on its maximum value defines the position of the *peak* of the wave packet. Upstream of the peak $k_i < 0$, while $k_i > 0$ downstream; hence, during its linear growth phase the packet envelope decays exponentially with distance from the peak. The structure near the peak is dominated by the most unstable normal mode. Second, the minimum (x/t) for which ω_i first becomes positive locates the *trailing edge* of the wave packet. As a consequence of (5.5), the trailing edge is dominated by short-wave, shallow modes (Pierrehumbert 1986). Finally, the maximum (x/t) for which ω_i returns to zero defines the *leading edge*. The leading edge speed is comparable to (though generally somewhat less than) the jet speed at $z = H$. Note that k_r does not change much between the peak and the leading edge. The phase speeds in Figure 5 are given relative to the ground; phase speeds get faster toward the leading edge, but remain compatible with a lower tropospheric steering level. Near the leading edge, phase lines propagate more slowly than the envelope of the packet, and so individual highs and lows appear to move toward the rear when viewed in the frame moving with the leading edge speed. Near the packet peak, however, phase propagation is faster than group propagation. This is somewhat sensitive to β^* , with large values giving greater retardation of phase propagation (Lin & Pierrehumbert 1993).

When the ray $(x/t) = 0$ lies in the cone of exponential growth, the flow is said to be *absolutely unstable*, because the packet leaves behind a disturbance which amplifies in situ. The two-layer model has weak absolute instability at zero surface wind (Merkine 1977), but the Charney model has no absolute instability. This was suggested by some results of Farrell (1982b), and first shown properly in Pierrehumbert (1986). Lin & Pierrehumbert (1993) solved the problem for realistic 2-D jets $u_0(y, z)$, and found results that are altogether consistent with the 1-D picture, particularly with regard to the absence of absolute instability for jets that are everywhere westerly. They also found that Ekman friction increases the speed of the trailing edge of the wave packet by suppressing the short-wave, shallow disturbances found there in the undamped case.

In Figure 6 we show an example of typical wave-packet evolution, simulated using the two-layer model (Swanson & Pierrehumbert 1994). This calculation has been carried through the nonlinear stage. At early times, the evolution is in accord with the expectations from the linear analysis. However, at late times, the trailing edge modes never reach appreciable amplitude, and the modes toward the leading edge assert their dominance.

The asymptotics discussed in Pierrehumbert (1986) guarantee that the trailing-edge modes are shallow. Because c_r gets larger and k_r gets smaller

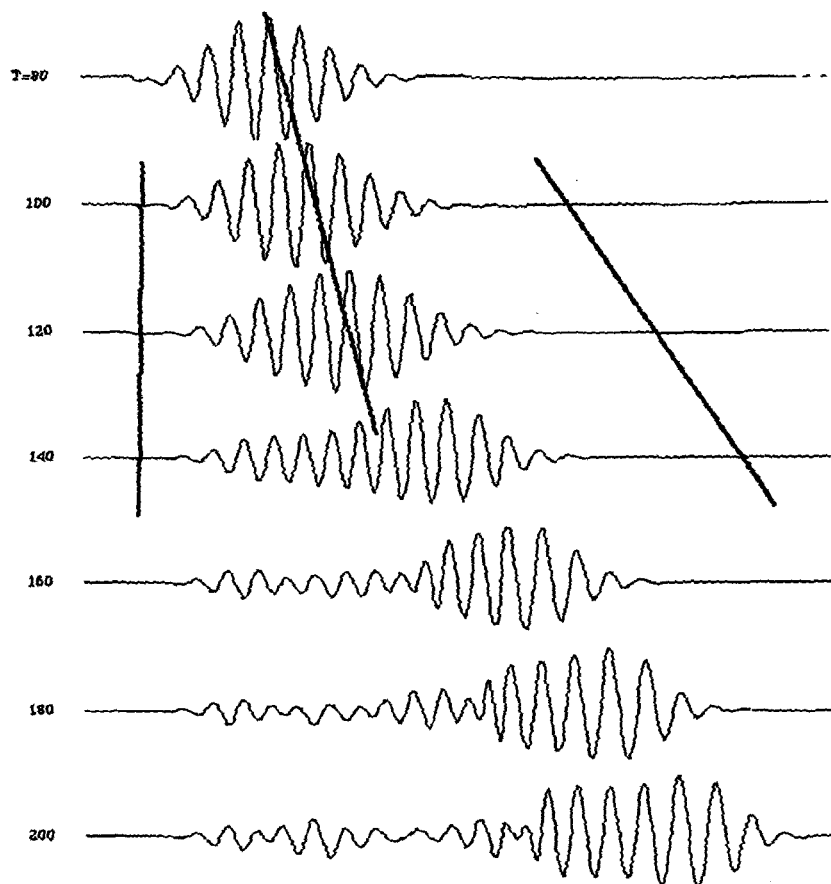


Figure 6 Linear and nonlinear stages of wave-packet evolution in a baroclinically unstable two-layer model. Curves indicate the upper-layer geopotential amplitude on the centerline of the channel, normalized to fixed maximum amplitude at each time frame. The rays indicating linear trailing edge, packet peak, and leading edge positions are also shown. From Swanson & Pierrehumbert (1994).

as the leading edge is approached, one expects the modes to become deeper there. These expectations are borne out by the mode structure computed by Farrell (1983) based on an approximate version of the dispersion relation. In Figure 7 we show a similar calculation for the case of Figure 5, carried out numerically using the full dispersion relation. The depth of the leading-edge modes in large measure accounts for their nonlinear dominance, for reasons to be discussed in Section 7. Fast leading-edge precursors of precisely the expected type have been found by Chang & Orlanski (1993) in idealized models and by Chang (1993) in the atmosphere.

When the flow is not absolutely unstable the *spatial instability* problem is well-posed. This problem deals with the response to a spatially localized temporally fluctuating source. Mathematically, we do this by seeking complex $k(\omega)$ for real ω , employing a selection criterion in the complex plane to distinguish between evanescent and amplifying solutions. Spatial instability was discussed by Thacker (1976) for the two-layer model and by Pierrehumbert (1986) for the Charney model, and a discussion of the implications for storm tracks and low frequency variability can be found in the latter.

Absolute instability plays an important role in the WKB theory for zonally inhomogeneous flow, allowing zonally localized eigenmodes in certain circumstances. The construction was first put forth in Pierrehumbert (1984) in the context of baroclinic instability, but it is quite general and has seen fruitful extensions and applications in other stability problems. These developments have been reviewed in Huerre &

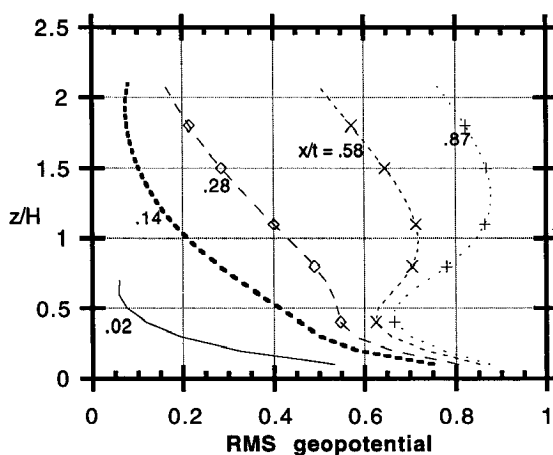


Figure 7 RMS geopotential profiles at various positions in a Charney model wave packet. Parameters as for Figure 5.

Monkiewicz (1990), and so we shall not dwell on them here. Applied to baroclinic jets, the theory produces some tantalizing similarities to observed 3-D storm track structure (Pierrehumbert 1985), but the lack (or weakness) of absolute instability in realistic profiles and the low equilibration amplitude of slow-moving, shallow trailing-edge disturbances makes this line of attack problematic.

The picture emerging from the preceding results is that an undisturbed baroclinic zone acts primarily as an amplifier of noise propagating in from upstream. The “gain” can be zonally quite inhomogeneous, as in the northern hemisphere winter, or it can be fairly homogeneous as in the southern hemisphere, owing to its weaker planetary waves. Either way, the central question comes down to the source of the excitation. This could be due to absolute instability, as in the local mode theory of Pierrehumbert (1984); Lin & Pierrehumbert (1993) showed that trailing-edge modes can amplify significantly even in the absence of absolute instability. Although the findings of Swanson & Pierrehumbert (1994) suggest that such shallow modes never attain very great amplitude, they could nonetheless conceivably act as triggers for more prominent developments. At present, there is no observational support for this scenario, however. There could be self-excitation due to upstream propagation of nonlinearly generated waves; Simmons & Hoskins (1979) and Swanson & Pierrehumbert (1994) find no self-seeding of this sort, but the situation could conceivably change if mountains or zonal flow inhomogeneities are taken into account. The excitation could come from the projection of fluctuating diabatic forcing due to convection and cloud-radiative effects on larger scales. This seems unlikely, because general circulation models with and without diurnal cycles and with a wide variety of cloud and convection schemes (all grossly deficient vis a vis the real atmosphere) reproduce a satisfactory level of synoptic eddy activity. Finally, the excitation could simply be due to debris that survives a previous life cycle and is reincarnated via amplification in the next undisturbed baroclinic zone it encounters. Because the jet flow on a sphere is reentrant, this process could sustain eddy activity indefinitely, in the form of a snake chasing its own tail. This picture has the important implication that suppression of eddies in one sector of the globe could affect eddy activity throughout an entire hemisphere.

Transient Growth Mechanisms (Sinusoidal in x)

The ideas behind nonmodal transient growth originate with studies by Orr (1907) and Case (1960) of unstratified shear flow. For nonmodal baroclinic transient growth the archetype is provided by the Eady problem in a semi-infinite domain: $\rho = \text{const.}$, $N = \text{const.}$, $u_0 = z$, and a rigid flat boundary at $z = 0$. With these assumptions $q' = \exp[i(kx + ly + (m - kt)z)]$ is an exact

solution to the perturbation PV equation. A corresponding solution for the streamfunction is

$$\psi' = \frac{e^{i(kx + ly + (m - kt)z)}}{l + (m - kt)^2} + A(t)e^{i(kx + ly + iz - t)}. \quad (6.3)$$

We have scaled horizontal lengths by $L = 1/(k^2 + l^2)^{1/2}$ and depths by $f_0 L/N$. The second term is a homogeneous solution added to satisfy the lower boundary condition, and its coefficient $A(t)$ becomes constant as $t \rightarrow \infty$ (Farrell 1984). The initial value problem for the general Eady profile was treated by Pedlosky (1964c), but the solution in the above form was first presented by Farrell (1984), who emphasized the implications of transient energy growth. The streamfunction ψ' amplifies from $O(1/m^2)$ at $t = 0$ to $O(1)$ at $t = m/k$, whereafter the first term decays. This is an example of transient growth due to evolution of the Poincaré length L_P , as discussed in Section 4. At large times an edge wave of amplitude $A(\infty)$ is left behind. The initial condition can be regarded as the superposition of an edge wave with a set of continuum modes; the edge wave is unmasked as the continuum modes decay.

Taking the limit $q_{0y} \rightarrow 0$ in (4.1–4.3) we find that pseudomomentum in this problem reduces to perturbation enstrophy, which is time invariant during the above evolution. The edge wave has zero pseudomomentum, and so can attain arbitrary amplitude without violating any conservation laws of the linearized system. For q_{0y} finite and positive, surface temperature variance reappears in the pseudomomentum, and there must be simultaneous amplification of enstrophy and temperature variance. The edge wave continues into an amplifying Charney mode for $q_{0y} > 0$ [cf (5.5)], which now gets the asymptotic projection and grows exponentially at long times. Like the edge wave, this mode has zero pseudomomentum, as indeed any amplifying normal mode must.

A zoo of other growth mechanisms is possible (Farrell 1982a, 1984, 1985; Farrell & Iouannou 1993). The problem (or opportunity, depending on one's point of view) is that the set of neutral modes for a given (k, l) is not orthogonal with respect to energy: The energy of a superposition is different from the sum of the energies of the individual modes. In consequence, the energy of the superposition can vacillate as the modes propagate and their relative phases change. Shepherd (1985) has noted, however, that the transient amplification can be less impressive when many k are superposed to yield a zonally localized initial condition.

Given the modest exponential growth rate of normal mode instabilities, one hardly expects the normal-mode structure to emerge unequivocally from arbitrary initial conditions during the few days typically available

before nonlinear equilibration sets in. Some sensitivity of the evolution to the form of the initial disturbance is inevitable, and this sensitivity can be elegantly expressed in terms of the optimal excitation construction (Farrell 1989). This just produces some fluctuations in the initial amplification rates. No compelling evidence that transient growth plays an *essential* role in generating the Earth's synoptic eddies has emerged, nor is there evidence that forcings such as fluctuating convection are needed to maintain eddy activity, or that nonlinear self-excitation of transient growth could maintain synoptic eddies in the absence of normal-mode instability. On the other hand, much evidence supports the picture of synoptic eddies as wave packets composed of growing normal modes, modified by progress of the nonlinear life cycle.

7. NONLINEAR EQUILIBRATION

With regard to climate, the chief goal of nonlinear equilibration studies is to determine the amplitude at which eddies stop growing, and to understand the attendant eddy-induced alterations of the basic state. A recurrent theme is the notion of "baroclinic adjustment," introduced by Stone (1978). The premise is that baroclinic instability acts to adjust the flow to a neutrally stable state. This supposition is advanced by way of analogy with convection, which (outside thermal boundary layers) adjusts the convecting fluid to a state of neutral static stability. A variety of pathways to baroclinic stabilization are available. Consider the neutralization of a flow with $q_{0y} > 0$, positive low-level shear, and a rigid bottom boundary. It is not sufficient to homogenize the meridional temperature gradient in a shallow layer near the boundary, for this would only create a region of negative q_{0y} immediately above the boundary. Ignoring horizontal shear and assuming $\rho = \rho_0 \exp(-z/H)$, the profile $\partial_z u_0 = \beta(N^2/f_0^2)H[(\rho_0/\rho) - 1]$ has $q_{0y} = 0$ and zero shear at the ground. This can be patched smoothly into a shear of magnitude a at a height z^* determined by $(\rho/\rho_0) = \beta^*/(1 + \beta^*)$, where β^* is the Charney number based on a . This is the neutralization construction proposed by Lindzen et al (1980). It can be realized quite naturally by synoptic eddies, via low-level mixing of temperature and PV, and the required mixing depth even scales similarly with the depth of normal-mode instabilities: For large β^* , z^* is $O(H/\beta^*)$ and becomes shallow like the modes. For small β^* deep mixing is required, and the eddies have the right vertical structure to provide it.

Gutowski (1985) noted that adjustment of N yields neutralization without the necessity of eliminating the temperature gradient. Specifically, the prescription $N^2/f_0^2 = \partial_z u_0 \rho / [A + (\rho_0 - \rho)\beta H]$, where A is an arbitrary constant, yields $q_{0y} = 0$. N -adjustment requires an increase in low-level static

stability, which is not available in quasi-geostrophic models but is possible in primitive equation models. Lindzen (1993) remarked that with meridional confinement to provide a lower bound on total horizontal wavenumber, the homogenized PV layer yields a stable flow even without mixing away the temperature gradient, provided the top of the layer is sufficiently high to decouple the upper level dynamics from the ground. We note further that a positive jump in N at the upper edge can be tolerated without instability, provided $\partial_z u_0 > 0$ there.

Either neutralization mechanism can be helped along by the barotropic governor effect. The matter of how adjustment occurs has been the subject of a great many numerical investigations, which can be divided among two chief categories.

Life-Cycle Studies

In *life-cycle studies* one tracks the evolution of a specified initial condition through its amplification and equilibration stage, and through the initial stages of the subsequent evolution. These problems are most cleanly addressed in the inviscid case, but some dissipation can usually be tolerated so long as it acts mainly on the perturbation and not on the zonal flow.

To set the stage, we shall describe a simple quasi-geostrophic equilibration calculation carried out for the Charney profile in a channel of width πL_d , with $\beta^* = 0.5$ and an initial small amplitude disturbance having wavenumber $kL_d = 1$. With $L_d = 1000$ km, one day of real time is three nondimensional time units. This example also serves as a summary of the rather intricate 3-D motion that releases the basic state potential energy without running afoul of angular momentum conservation. Results are shown in Figure 8.

Purely horizontal trajectories could mediate the necessary heat flux, but the constraints of quasi-geostrophy make some vertical motion inevitable. If the trajectories lay in the plane of the basic state isentropic surface, no potential temperature deviation θ' would be generated, and hence no heat flux; if the trajectories were *steeper* than the initial surface, then the heat flux would have the wrong sign to release energy. Only trajectories with slopes between horizontal and the slope of the mean isentropic surface can release the potential energy. This is the classical “wedge of instability” argument (see Pedlosky 1979). The heat-transporting trajectories are poleward/upward and equatorward/downward for the customary terrestrial tropospheric temperature distribution, and this is precisely the pattern seen in Figure 8, where we have mapped the vertical velocity onto the 280°K isentropic surface. Through condensation in ascending, adiabatically cooling trajectories, the vertical velocity pattern provides the

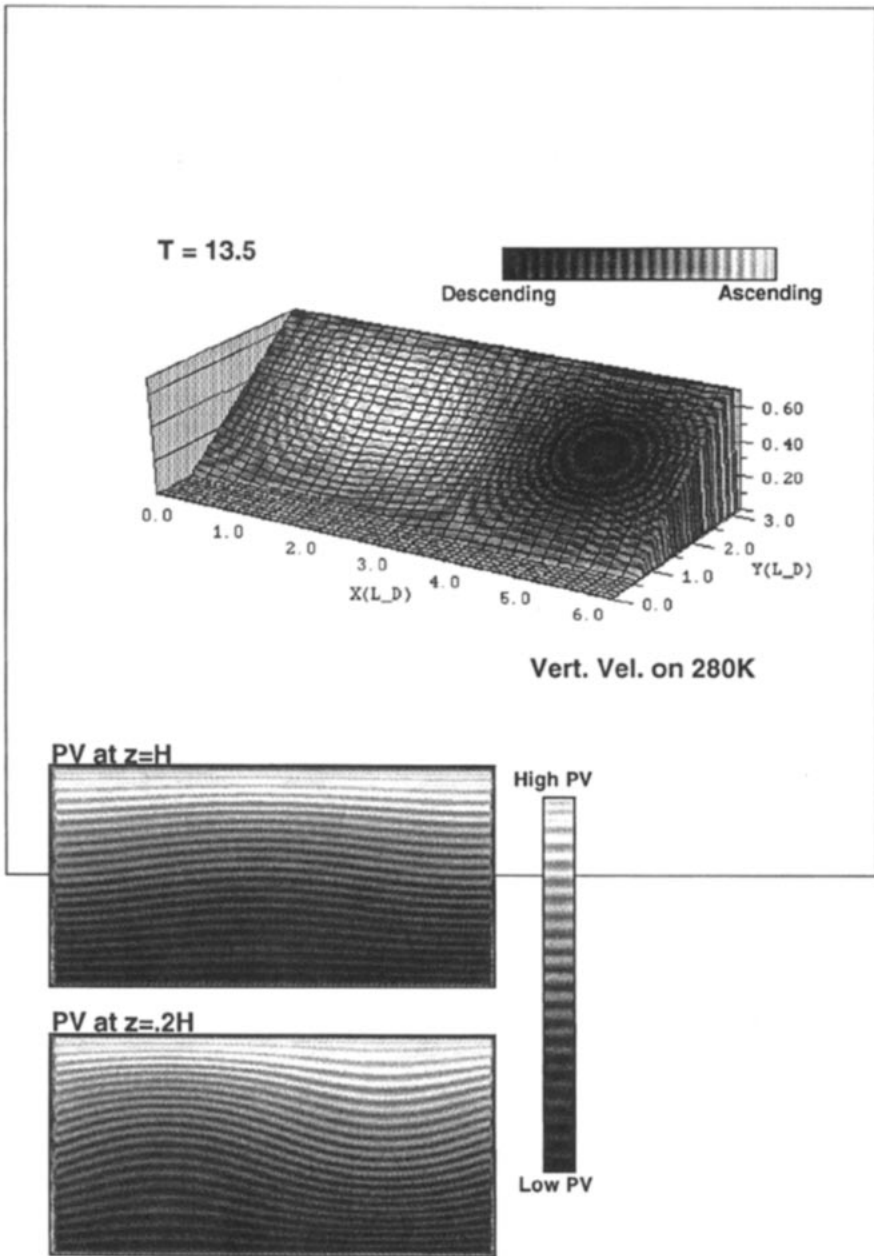


Figure 8 Nonlinear evolution for the Charney model in a channel. The top panel for each time shows the 280°K isentropic surface shaded according to vertical velocity. Bottom panels show potential vorticity at $z = 0.2H$ and $z = H$.

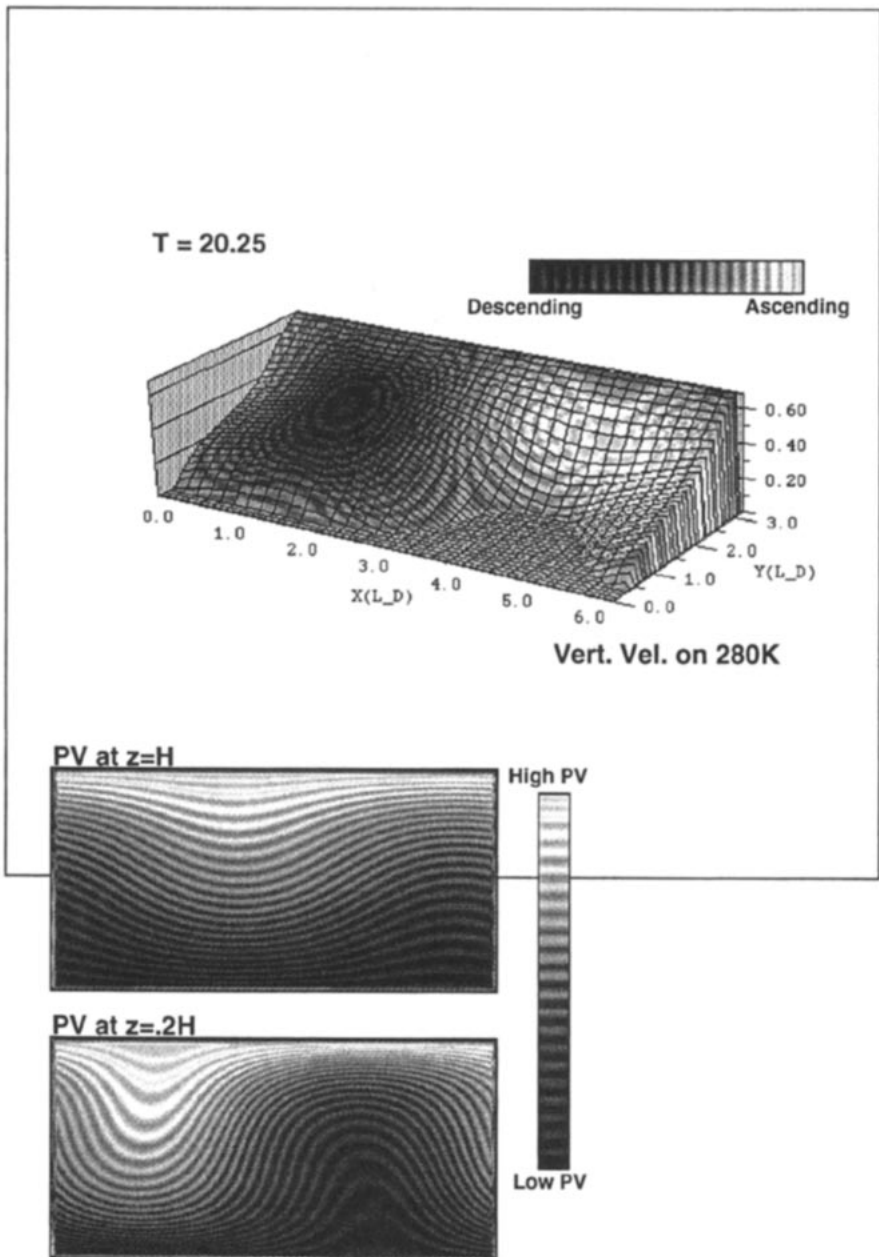


Figure 8—(continued)

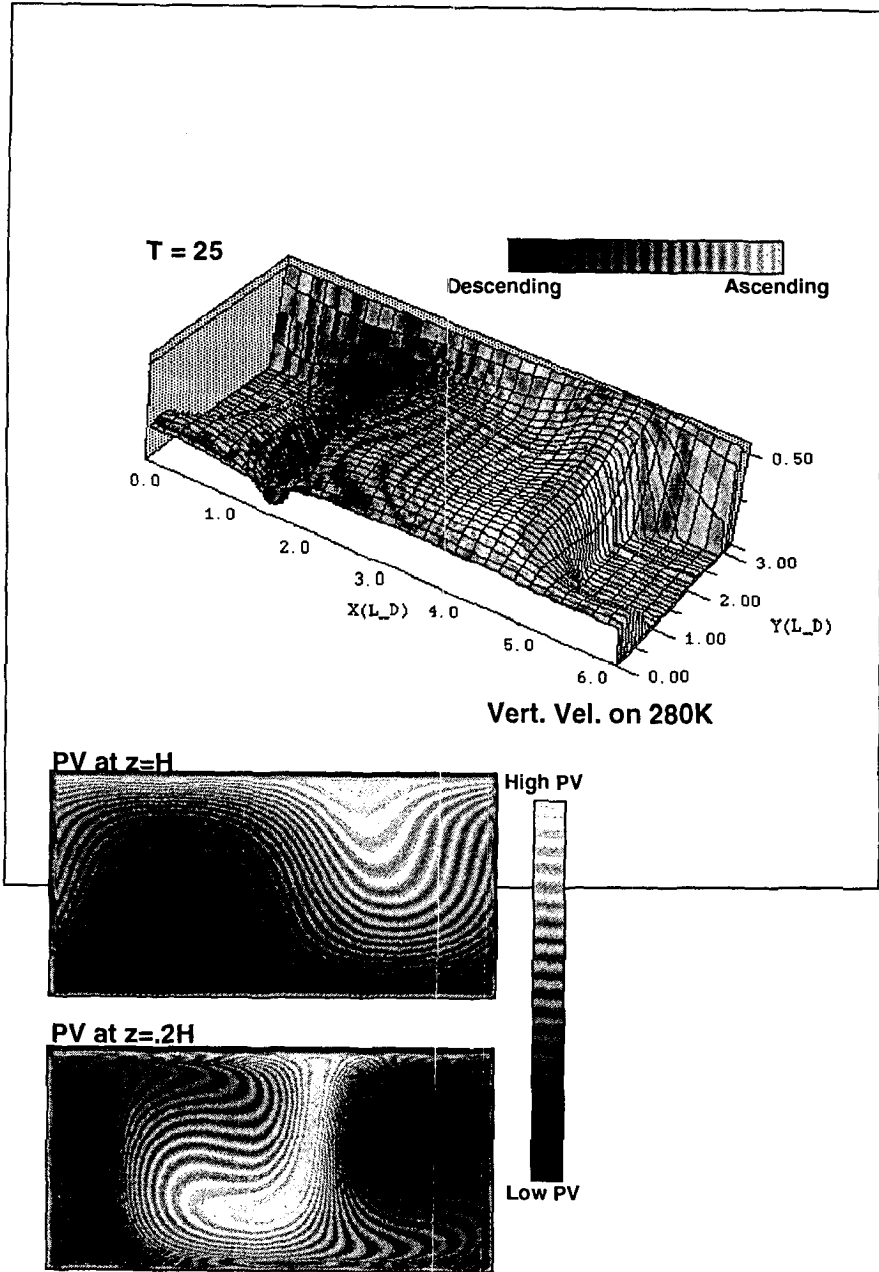


Figure 8—(continued)

link to the hydrological cycle; it implies a potent drying mechanism for atmospheres with a condensible component (Yang & Pierrehumbert 1994).

Cold polar air simply “slumps” down into the subtropics, with adjacent warm subtropical air rushing polewards to take its place. In the end state the zonal-mean temperature gradients at low levels have been expelled to the channel boundaries, much as high Rayleigh number laboratory convection expels gradients to the upper and lower thermal boundary layers. However, the end-state temperature is not zonally homogeneous. Cold and warm air have been rearranged to form consecutive zonal blobs separated by strong gradients. In quasi-geostrophic theory, PV mixing occurs on *horizontal* planes, not on *isentropic* surfaces, and so the PV patterns are shown at fixed z in Figure 8. At low levels, the zonal mean PV becomes meridionally homogenized, though similarly to temperature it remains zonally inhomogeneous in the end state.

At upper levels PV mixing is not so pronounced. The basic reason is that $u_0 - c$ is large aloft, so that the streamline displacements in the reference frame comoving with the wave are small unless the wave attains very great amplitude. The issue of PV mixing in more general flows is tied up with the following matters, which are all part and parcel of the same thing: critical levels, closed streamlines, heteroclinic structures, mixing by chaotic advection, Chirikov resonance overlap, and “wave breaking” in the sense of McIntyre & Palmer (1983, 1985). The connections can be understood with reference to the streamline geometry for a jet flow perturbed by a traveling wave (viewed in the frame comoving with the wave). An example is shown in Figure 9. One finds closed eddies in the vicinity of the critical lines $u_0(y) = c$. For fixed c their meridional thickness increases with perturbation amplitude, and the eddies come to span the central portion of the channel when c approaches the speed of the jet maximum. Large eddies can transport potential vorticity or other tracers a long way, and fluid trapped in the recirculating regions will wrap up and ultimately homogenize. On the other hand, there is little mixing near the jet core when the critical level is far out on the flanks. These considerations lead to a general interpretation for why instability is associated with critical levels: The resulting flow structures, on purely kinematic grounds, are favorable to the lateral mixing that releases energy. This reasoning is robust, and carries over into the highly nonlinear regime so long as the motions involved have a reasonably well-defined phase speed. For the quasi-geostrophic life cycle discussed above, the required mixing is due to the weak mean winds near the ground, and the low-level streamline pattern is like the case (4,1) in Figure 9. Warn & Gauthier (1989) presented an illuminating treatment of this sort of mixing in a weakly nonlinear model.

The streamlines anchored in stagnation points are, in dynamical systems

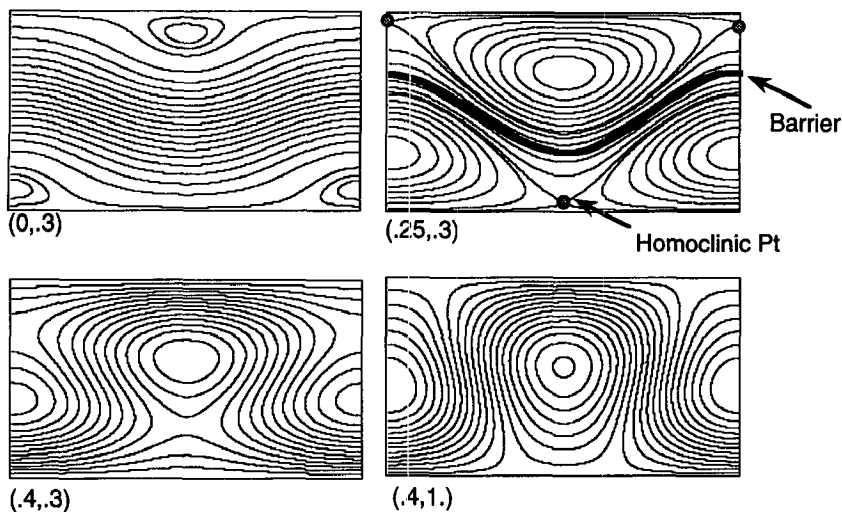


Figure 9 Streamlines of $\psi = cy - \tanh(\frac{1}{2}y) + \epsilon \cos(x) \cos(\frac{1}{2}y)$ for the indicated values of (c, ϵ) . The locus of the transport barrier, and a homoclinic point P are marked for $(.25, .3)$.

parlance, “homoclinic cycles,” and it is well understood that such things lead to chaotic mixing when perturbed by transients [see Ottino (1989) for a general perspective, and Pierrehumbert (1991a,b, 1992) and Pierrehumbert & Yang (1993) for an atmospheric point of view]. Conversely, the open-streamline region snaking between the gyres tends to form a barrier to transport (Pierrehumbert 1991b, 1992; del-Castillo-Negrete & Morrison 1993 and references therein). The Chirikov resonance “criterion” is little more than the rather vague statement that global chaos (in the present case strong mixing across the jet barrier) ensues when the mixing zones surrounding the upper and lower homoclinic streamlines grow thick enough to overlap. Obviously, overlap is easier in a case like $(.25, .3)$ in Figure 9 where the homoclinic streamlines are close than in a case like $(0, .3)$. In any event, chaotic advection leads to (macroscopically) irreversible PV mixing. This is essentially the “wave breaking” referred to in McIntyre & Palmer (1983); though they deal with Rossby waves rather than synoptic eddies, the implications of streamline geometry are identical.

While quasi-geostrophic theory captures the essentials of baroclinic instability throughout most of the troposphere, it is grossly inadequate at representing the PV mixing across the tropopause. Quasi-geostrophic eddies do not mix potential vorticity along isentropic surfaces, and this becomes a serious shortcoming aloft because “middleworld” surfaces like $\theta = 315^\circ\text{K}$ cross the tropopause at a steep angle (Hoskins 1991). As indi-

cated in Figure 10, synoptic transients would mix high- N , high-PV stratospheric air into the troposphere if properly represented. In a full general circulation model, Pierrehumbert & Yang (1993) found free mixing across the tropopause on the 315°K surface. Such mixing almost certainly contributes significantly to the tropospheric and stratospheric PV budgets. The structures resulting from the intrusion of stratospheric air into the troposphere are generally known as “tropopause folds.”

For the inviscid quasi-geostrophic system the net enstrophy at each z is an exact nonlinear invariant. Assuming initial perturbation enstrophy to be negligible it follows that

$$\int \overline{q'^2} dy = \int \overline{q_0'^2} dy = \int \overline{q_i'^2} dy + \int \overline{q'^2} dy, \quad (7.1)$$

where overbars denote x -averages and the subscript “f” denotes the final state. Hence, eddy enstrophy is rigorously bounded by the difference between the enstrophies of the initial and adjusted zonal mean flow. A similar argument was put forth by Schoeberl & Lindzen (1984) for barotropic adjustment. The same considerations apply *mutatis mutandum* to the surface temperature variance, when there is no Ekman damping. These bounds can sometimes be improved by use of an Arnol’d invariant (Shepherd 1988, 1989), when there is a suitable “nearby” zonal flow that is nonlinearly stable. Let’s suppose that the eddies are confined to a meridional strip of length L_y , either through channel walls, jet structure, or some other mechanism. More broadly, L_y can be considered to be a mixing length, corresponding to the meridional displacements caused by

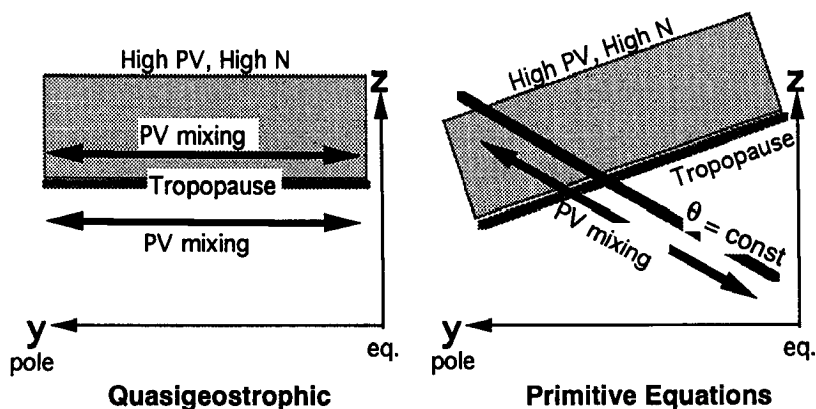


Figure 10 Mixing of PV across the tropopause: quasigeostrophic vs primitive equation behavior.

the eddies. Then, the maximum eddy enstrophy is attained if $\bar{q}_f$ is obtained by homogenizing q_0 over L_y . For constant q_{0y} , the mean perturbation enstrophy is

$$\frac{1}{L_y} \int \bar{q}^2 dy = \frac{1}{3} (q_{0y})^2 L_y^2, \quad (7.2)$$

which becomes unbounded as $L_y \rightarrow \infty$. There is nothing very mysterious here: Perturbation q' is created by displacing the initial PV in y , and the greater the range of PV that can be tapped, the greater the perturbation enstrophy that can be created. This estimate underscores the importance of the mechanisms controlling the meridional scale of the synoptic eddies. In life-cycle studies, the meridional scale tends to be set by the width of the preexisting jet.

Simmons & Hoskins (1978) find equilibration due to low-level mixing such as described above. Gall (1976b) and Gutowski et al (1989), on the other hand, find a greater role for N -adjustment in stabilization. All three studies were conducted using the primitive equations, and therefore are not subject to the limitations on adjustment imposed by quasi-geostrophy, nor are there any obvious differences in the overall parameter range treated. The discrepancy in adjustment may be due to subtle differences in treatment of dissipation; in light of Section 5, it seems possible that Ekman friction could affect the late stage of the equilibration, after the shear has been weakened somewhat. Not all bases have been covered with regard to exploring such frictional effects. It is also noteworthy that the latter two studies used a wave-mean-flow model truncated to a single zonal wave-number, whereas Simmons & Hoskins (1978) employed a full primitive equation model. Nakamura & Held (1989) and Garner et al (1992) show N -adjustment in a primitive equation model which by construction precludes shear-adjustment, but argue that N -adjustment should predominate only for eddies that are highly elongated in y . This is not the case in the studies of either Gall (1976b) or Gutowski et al (1989). Currently, both circumstances under which N -adjustment can dominate, and the role of dissipative effects, remain obscure.

Nonlinear effects do remove the two major shortcomings of linear eigenmodes vis à vis the observations, namely that the predicted modes are too shallow and that there is insufficient scale selectivity. Gall (1976b) found that longer waves equilibrate at greater amplitude than the linearly most unstable mode. The longer wave has more vertical penetration even in linear theory (recall Figure 4), and at equilibration the upper maximum comes to dominate the secondary surface maximum, owing to the vertical wave radiation effects alluded to previously. These effects were seen also

by Simmons & Hoskins (1978). Deeper modes equilibrate at higher energy in part because they can tap more of the energy and enstrophy of the mean flow, but scaling considerations also show that deep modes tend to have a greater streamfunction amplitude for a given PV amplitude. With depth scale D and horizontal scale no smaller than the corresponding radius of deformation, scaling of (3.1b) implies $\psi' \approx [(N^2/f_0^2)D^2]q'$, so long as $D \leq O(H)$. While these arguments provide a handle on low-level equilibration, the theoretical determination of amplitude aloft following equilibration is still an unsettled matter.

Following equilibration, the typical behavior in the strongly nonlinear case, at least for sinusoidal global perturbations, is *barotropic decay*. When the disturbance stops growing, it can be thought of as a superposition of equivalent-barotropic neutral modes. These can be continuum modes that decay algebraically through destructive interference; such behavior is more commonly interpreted in terms of Rossby waves that radiate meridionally into critical levels where they are absorbed, but the two viewpoints are mathematically equivalent. Either way, the evolution results in a barotropic energy sink term in (3.5), and feeds eddy pseudomomentum back into the barotropic zonal flow. Some of the projection could also go into modes that radiate away vertically without feeding into the barotropic flow in the troposphere, or into trapped modes which persist for all time, apart from dissipative effects. Simmons & Hoskins (1978) find barotropic decay, and Feldstein & Held (1989) discussed the criteria for its appearance in a family of simplified models. Barotropic decay is a kind of upscale energy cascade and could presumably feed into planetary waves (as in the barotropic example in Pierrehumbert 1991a) instead of the zonal flow. The asymmetry (baroclinic growth followed by barotropic decay) is important because it means that each individual life cycle yields a net heat flux even in the absence of diabatic forcing. This may be contrasted with symmetric baroclinic growth followed by baroclinic decay, as found in inviscid weakly nonlinear models (Pedlosky 1970) and other situations where barotropic decay is inhibited (Feldstein & Held 1989). Without the asymmetric life cycle, baroclinic eddies in the conservative system would act only as a transient repository of energy, and on the average would transport no heat. The character of the barotropic decay stage is greatly influenced by barotropic jets generated nonlinearly in the course of the evolution (Nakamura 1993b).

The above picture prevails for life cycles proceeding from zonally sinusoidal initial disturbances, and although our understanding is not entirely complete, it is nearly so. Recent simulations suggest that the life cycles of zonally localized packets may be quite different from the global sinusoidal

case (Swanson & Pierrehumbert 1994). The equilibration mechanism for packets is similar, and deep leading-edge modes attain the largest equilibrated amplitudes for the same reason that long waves win out in the sinusoidal case. However, the barotropic decay stage is greatly affected because the nonlinearly generated barotropic jets appear upstream of the energy-releasing eddies, rather than being co-located. In this context, it is notable that Chang & Orlanski (1993), in a study of eddies amplifying downstream of a localized forcing, find no inherent downstream limitation of storm track due to barotropic decay.

Equilibrium Studies

In *equilibrium studies* the zonal flow is maintained by differential heating, and one investigates the long-term statistical behavior of the fluctuating system, which is presumably independent of initial conditions. This subject deserves an article in its own right, under the heading of “baroclinic turbulence.” Space permits us only to provide a few pointers into the literature. Few, if any, of the important issues have been definitively resolved. Central among these is the matter of why synoptic eddies on the Earth are quite effective at mixing away PV gradients on isentropic surfaces (Sun & Lindzen 1994, Pierrehumbert & Yang 1993) but not very effective at eliminating the low-level temperature gradient.

One category of studies (Held 1973a; Zhou & Stone 1993a,b) deals with the scaling of heat fluxes with temperature gradient, and the adjustment of the temperature pattern by baroclinic eddies. In these studies, the meridional width of the active domain is on the order of a radius of deformation, as in the present-day terrestrial climate.

A second category deals with situations where the baroclinicity extends over many radii of deformation in y (Panetta & Held 1989, Panetta 1993). In a meridionally unbounded domain with no basic-state horizontal shear, amplifying waves with meridional wavenumber $l = 0$ are exact nonlinear solutions of the quasi-geostrophic system, and therefore never equilibrate, though they succumb eventually to a secondary instability (Pedlosky 1975). It seems likely that similar instabilities exist for sufficiently elongated eddies; the general tendency to isotropy could be thought of as due to secondary rollup of elongated eddies in much the way that an elongated elliptical vortex patch in the 2-D Euler equations fissions or rolls up (cf Dritschel 1986). Once strong barotropic jets appear, they can further influence eddy anisotropy by shearing out and “eating” elongated eddies. In fact, the wide domain limit doesn’t always yield a meridionally homogeneous eddy field; typically there is symmetry breaking due to spontaneous zonal jet formation, and the jets organize the eddies into multiple storm tracks. The equilibrium zonal mean flow is highly unstable, rather

than being adjusted to marginal stability. All of this is suggestive of the banded structure of the gas-giant atmospheres.

Finally, it has been found that localized wave packets appear spontaneously in the equilibrium problem (Lee & Held 1993). This is a very important result, and the implications are only just beginning to be digested.

8. CONCLUDING REMARKS

Baroclinic instability is unambiguously successful at explaining why differentially heated rotating planets spontaneously generate transient eddies rather than settling into a local radiative equilibrium. It also tells us under what condition a flow is stable, and thus gives an impression of what kinds of state the system might be relaxing toward. It identifies the kinds of motions that can release potential energy, and has yielded useful insights as to the interplay of baroclinic eddies and static stability, the dominant spatial scales involved, and the vertical structure of the eddies and their heat fluxes. Further, all this works better than one has any right to expect at describing synoptic eddies appearing in forced-dissipated equilibrium systems: the Earth's atmosphere, general circulation models, and simplified models.

The essence of the instability resides in the coupling of a pair of vertically separated potential vorticity disturbances, which can maintain phase propagation at a speed supporting a critical level in the flow. The critical level, in turn, supports effective lateral mixing even for an initially small amplitude disturbance. When there is a rigid boundary, the surface potential temperature perturbation can take on the role of one of the potential vorticity disturbances. Not just any critical level will do. Special conditions on the background potential vorticity distribution near the critical level must be met in order for the two disturbances to remain together as they propagate. In the absence of these conditions, the disturbances eventually separate, and mixing proceeds only for a finite time. The essence of non-linear equilibration also comes down to a matter of mixing depleting the environmental gradients, and the configuration of the mixing zones can be tied to the separatrix structure of the net streamline pattern. As the mixing proceeds the special background potential vorticity conditions are eliminated, whereafter the vertically separated perturbations decouple and go their own separate ways.

Though the best current estimates indicate that the Earth's atmosphere permits vigorous exponential amplification of disturbances, the terrestrial baroclinic jets have—at best—only a very weak absolute instability. Evidence from simulation and observations suggests that the debris left behind

by wave packets is too weak to play an appreciable role in the synoptic variability. Baroclinic jets thus constitute an “open system” in the sense of Huerre (1987). Like boundary layers or viscous pipe-flow, storm tracks have “receptivity” to imposed perturbations, and therefore, the nature of the eddies cannot be divorced from the nature of the excitation. A better understanding of the nature of the excitation in planetary atmospheres is clearly needed.

Baroclinic instability theory is a well-honed tool, whetted for application to other planets. Mars presents an especially promising opportunity: It is like the Earth in rotating rapidly and having a rigid surface, but unlike the Earth in being dry and strongly damped thermally. It is an outstanding laboratory for examination of the interplay of baroclinic eddies and static stability, in the absence of the complicating effects of moisture. The loss of the *Mars Explorer* satellite is truly tragic in this regard, and it is greatly to be hoped that a replacement mission can someday be mounted. Jupiter and the other gas giants exemplify a system in which instability takes place without benefit of a rigid boundary, and perhaps even one with no normal-mode instability at all. They also present the archetype of behavior for planets large compared to a radius of deformation. Overall, the determination of static stability N looms as a key issue, because the radius of deformation is proportional to N . Why doesn't Earth evolve towards a low- N state, nearly isothermal in the horizontal, bearing a multiple jet circulation like that of Jupiter? Even for our own Earth, much ground remains to be covered with regard to testing of predictions against the seasonal cycle and against the differences between the northern and southern hemispheres.

Baroclinic instability is far from a closed book. Even in very classical areas, there is a resurgence of interest in the physical relevance of Eady-type models (Lindzen 1994). Further work is needed on the effects of zonal inhomogeneity (and especially the role of diffluence in terminating storm tracks). Much remains to be learned about upper level equilibration amplitudes and fluxes of potential vorticity, as well as about moisture transport, latent heat effects, and cloud feedbacks. Moreover, though we have emphasized synoptic eddies in this review there is a possible role for planetary-scale baroclinic instability in generating low-frequency variability.

What is of most interest at this point is “baroclinic turbulence,” not baroclinic instability. Scaling laws are of little help, because it is the energy-injecting eddies rather than the inertial range that is of primary interest. On the other hand, the problem is very amenable to computation, though it has proved resistant to theoretical understanding at a deep, quantitative level. The central questions revolve around the matter of why the system

spontaneously generates “life cycles” and the form of the excitation for each new life cycle, and around the emergence of spatially localized wave packets and the amplification/“healing” sequence. This is wrapped up with the poorly characterized radiative-convective processes that restore the low-level temperature gradients and which help prevent in Nature the extreme temperature fluctuations seen in undamped life cycles. The fact that life-cycle theory does so well is in itself a clue: Unstable eigenmodes for convection, by contrast, are essentially irrelevant to the developed turbulent system. It suggests a kind of intermittency in baroclinic energy release in the terrestrial regime.

As we come to the close of the Millennium, it can be said with some fairness that our times have not done too badly at coming to an understanding of the fundamental means by which the climates of rotating planets operate. Linear instability theory and the emergent nonlinear life cycles have been pushed pretty far already in this endeavor, and perhaps it is better not to attempt to push them any farther. We have some optimism that the coming times will see the appearance of tools more ideally suited to the treatment of baroclinic turbulence.

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